On continuous-time models of GD and SGD with momentum

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I. BEA for GD with momentum

1. Setup

The GD with momentum update is written as:

$$x_{t+1} = x_t - \eta \nabla f(x_t) + \mu(x_t - x_{t-1}), \qquad t \geq 0, \qquad x_0 = x_{-1}.$$

We will use the following ansatz from Backward Error Analysis:

$$\dot{x} = g(x) + \sum_{p=1}^{\infty} \eta^p g_p(x), \qquad g(x) = -\frac{\nabla f(x)}{1-\mu}.$$

We fix $t \geq 1$ and assume $x_t = x(0)$. The main idea to choose the functions g_p so that the following two expressions match under the principle of ignoring terms containing $\nabla^p f$ for $p \geq 3$:

$$x(\eta) + \mu x(-\eta) \text{ with } (1+\mu)x_t - \eta \nabla f(x_t).$$

To do the matching, we use a forward and backward Taylor expansion, inspired by Lyu et al 2025:

$$x(\eta) + \mu x(-\eta) = \sum_{p=0}^{\infty} \frac{\eta^p}{p!} \left(1 + \mu(-1)^p\right) x^{(p)}(0).$$

We need

$$(\mu+1)x_t + \eta(1-\mu)\left(-\frac{\nabla f(x_t)}{1-\mu}\right) + \sum_{n=1}^{\infty}(1-\mu)\eta^{n+1} g_n(x_t) + \sum_{k=2}^{\infty} \frac{\eta^k \left(1+\mu(-1)^k\right)}{k!} x^{(k)}(x_t) = (\mu+1)x_t - \eta \nabla f(x_t),$$

i.e. we need

$$\sum_{n=1}^{\infty}(1-\mu)\eta^{n+1} g_n + \sum_{k=2}^{\infty} \frac{\eta^k \left(1+\mu(-1)^k\right)}{k!} x^{(k)} = 0. \tag{1}$$

2. Deriving a recursion

Lemma A.3 from Rosca et al. 2023:

$$x^{(k)} = \sum_{m \geq 0} \eta^m \frac{d^{k-2}}{dt^{k-2}} \sum_{i+j=m} \nabla_x g_j \, g_i, \qquad k \geq 2.$$

Let

$$\frac{d^{k-2}}{dt^{k-2}} \sum_{i+j=m} \nabla_x g_j g_i = \sum_p \eta^p A^p_{k,m}.$$

Substituting this into (1),

$$g_n = -\sum_{k=2}^{n+1} \frac{1+\mu(-1)^k}{(1-\mu)k!} \sum_{m=0}^{n+1-k} A_{k,m}^{n+1-k-m}$$

Functional structure proven by induction:

$$g_n = c_n(\nabla_x g)^n g + C(\nabla_x^2 g),$$

where

$$c_0 = 1, \qquad c_1 = -\frac{\mu+1}{2(1-\mu)}, \qquad c_2 = \frac{(\mu+1)^2}{2(1-\mu)^2} - \frac{1}{6}.$$

From Lemma A.4 from Rosca et al.,

$$A_{k,m}^p = \sum_{i+j=m} \sum_{\substack{l_1+\cdots+l_{k-2}=p \\ l_i \geq 0}} c_{l_1} \cdots c_{l_{k-2}} (\nabla_x g)^{p+m+k-1} g + C(\nabla_x^2 g).$$

Similarly to A.4,

$$c_n = -\sum_{k=2}^{n+1} \frac{1+\mu(-1)^k}{1-\mu} \frac{1}{k!} \sum_{l_1+\cdots+l_k=n+1-k} c_{l_1} \cdots c_{l_k}.$$

3. Solving the recursion

Similarly to Lemma A.5, the generating function is

$$c(x) = 1 - \frac{1}{x} \sum_{k=2}^{\infty} \frac{1+\mu(-1)^k}{1-\mu} \frac{x^k}{k!} c(x)^k.$$

This becomes

$$= 1 - \frac{1}{x} \left(\sum_{k=2}^{\infty} \frac{x^k}{k!} c(x)^k + \sum_{\substack{k=2 \\ 2|k}}^{\infty} \frac{2\mu}{1-\mu} \frac{x^k}{k!} c(x)^k \right),$$

which simplifies to

$$= 1 - \frac{1}{x} \left(e^{xc(x)} - 1 - xc(x) + \frac{2\mu}{1-\mu} \cdot \frac{e^{xc(x)} + e^{-xc(x)} - 2}{2} \right).$$

Hence

$$(1-\mu)x + \mu + 1 = e^{xc(x)} + \mu e^{-xc(x)}.$$

Equivalently,

$$e^{2xc(x)} - \big((1-\mu)x + \mu + 1\big) e^{xc(x)} + \mu = 0.$$

This quadratic has two positive roots for $\mu > 0$ and $x > -\frac{1-\sqrt{\mu}}{1+\sqrt{\mu}}$. We choose the larger root since this agrees with the choice for $\mu = 0$. This is also necessary, because if the two roots are $y_\pm(x)$, we have $y_+(0) = 1$, $y_-(0) = \mu$ and since $c(0) = c_0 = 1$, we have $\lim_{x\to 0} \frac{\log y_\pm(x)}{x} = 1$, which for the smaller root having $y_-(0) = \mu$ doesn't hold.

$$\Rightarrow \quad c(x) = \frac{1}{x} \log \left(\frac{(1-\mu)x + \mu + 1 + \sqrt{(1-\mu)^2 x^2 + 2(1-\mu)(\mu+1)x + (1-\mu)^2}}{2} \right),$$

with radius of convergence

$$R(\mu) = \frac{1-\sqrt{\mu}}{1+\sqrt{\mu}}.$$

4. Principal flow for GD with momentum

$$\dot{x} = \sum_{p=0}^{\infty} c_p \eta^p \left(-\frac{\nabla^2 f}{1-\mu}\right)^p \left(-\frac{\nabla f}{1-\mu}\right) = \sum_{p=0}^{\infty} \frac{(-1)^{p+1}}{(1-\mu)^{p+1}} c_p \eta^p (\nabla^2 f)^p \nabla f,$$

where

$$c(x) = \sum_{p=0}^{\infty} c_p x^p,$$

and truncating up to order n gives, modulo terms involving $\nabla^p f,\ p \geq 3$,

$$\|X((k+1)\eta) + \mu X((k-1)\eta) - (\mu+1)X(k\eta) + \eta \nabla f(X(k\eta))\| \leq O(\eta^{n+2}).$$

If

$$\nabla^2 f = U \Lambda U^\top,$$

then

$$\boxed{\dot{x} = -\frac{1}{1-\mu}\, U\, c\!\left(-\frac{\eta\Lambda}{1-\mu}\right) U^\top \nabla f,}$$

for stepsize

$$\eta < \frac{(1-\sqrt{\mu})^2}{\|\nabla^2 f\|},$$

(the argument of c is smaller than $R(\mu)$.) This can be rewritten as:

$$\boxed{\dot{x} = \frac{U(x)\, \log\big(f(\mu I, \eta\Lambda(x))\big)\, U(x)^T}{\eta\Lambda(x)} \nabla f(x)},$$

where

$$f(\mu, x) = \frac{\mu + 1 - x + \sqrt{(\mu+1-x)^2 - 4\mu}}{2}$$

5. Error guarantees

In this section assume that the derivatives of f are uniformly bounded (e.g. $f \in C^2$), in particular let $\|\nabla^2 f\|_\infty \leq M$..

Let

$$\varepsilon_k = x(k\eta) - x_k,$$

and let the initial condition of the ODE be

$$x(0) = x_0.$$

Notice that we only have $\varepsilon_1 = O(\eta)$. Indeed, since the derivatives of f are bounded and $x^{(k)}(0)$ depend only on them,

$$x(\eta) - x_1 = \sum_{k=0}^{\infty} \frac{\eta^k}{k!} x^{(k)}(0) - (x_0 - \eta \nabla f(x_0)) = \eta(\nabla f(x_0) - \frac{\nabla f(x_0)}{1-\mu}) + O(\eta^2) = O(\eta),$$

since there is a mismatch between the terms of order η. Let $\|\varepsilon_1\| \leq C_1 \eta$.

We will consider the case when the series is truncated up to order n. Then we have the following guarantee.

Claim. Let $T \geq 0$ be fixed; then $\sup_{0 \leq \eta \leq \frac{T}{\eta}} \|\varepsilon_n\| = O(\eta)$.

Proof. Due to BEA, from the matching itself we have

$$\|X((k+1)\eta) + \mu X((k-1)\eta) - (\mu+1)X(k\eta) + \eta \nabla f(X(k\eta))\| \leq O(\eta^{n+2}).$$

Hence

$$\|\varepsilon_{k+1} + \mu\varepsilon_{k-1} + x_{k+1} + \mu x_{k-1} - (\mu+1)x_k + \eta \nabla f(X(k\eta))\| \leq O(\eta^{n+2}).$$

Therefore

$$\left\|\varepsilon_{k+1} + \mu\varepsilon_{k-1} - (\mu+1)\varepsilon_k + \eta\big(\nabla f(X(k\eta)) - \nabla f(x_k)\big)\right\| \leq O(\eta^{n+2}).$$

Hence

$$\|\varepsilon_{k+1} + \mu\varepsilon_{k-1} - (\mu+1)\varepsilon_k\| \leq O(\eta^{n+2}) + \eta \left\|\nabla f(X(k\eta)) - \nabla f(x_k)\right\|.$$

Due to mean value theorem and the boundedness of the Hessian,

$$\|\nabla f(X(k\eta)) - \nabla f(x_k)\| \leq M\|\varepsilon_k\|,$$

we get

$$\|\varepsilon_{k+1} + \mu\varepsilon_{k-1} - (\mu+1)\varepsilon_k\| \leq O(\eta^{n+2}) + \eta M\|\varepsilon_k\|.$$

Thus if $\delta_k := \epsilon_k - \epsilon_{k-1}$, we have:

$$\|\delta_{k+1}\| \leq \mu\|\delta_k\| + \eta M\|\epsilon_k\| + C_r\eta^{n+2}.$$

By iterating this, we get:

$$\|\delta_m\| \leq C_1\eta\mu^{m-1} + \eta M \sum_{j=1}^{m-1} \mu^{m-1-j}\|\epsilon_j\| + C_r\eta^{n+2} \sum_{j=1}^{m-1} \mu^{m-1-j}.$$

Summing these, we get:

$$\|\epsilon_k\| \leq C_1\eta \sum_{m=1}^{k} \mu^{m-1} + \eta M \sum_{m=2}^{k} \sum_{j=1}^{m-1} \mu^{m-1-j}\|\epsilon_j\| + C_r\eta^{n+2} \sum_{m=2}^{k} \sum_{j=1}^{m-1} \mu^{m-1-j}$$

Note that

$$\sum_{m=2}^{k} \sum_{j=1}^{m-1} \mu^{m-1-j}\|\epsilon_j\| = \sum_{j=1}^{k-1} \|\epsilon_j\| \sum_{m=j+1}^{k} \mu^{m-1-j} \leq \frac{1}{1-\mu} \sum_{j=1}^{k-1} \|\epsilon_j\| \text{ and } \sum_{m=2}^{k} \sum_{j=1}^{m-1} \mu^{m-1-j} \leq \frac{k}{1-\mu}.$$

Combining these (and using $k\eta \leq T$):

$$\|\epsilon_k\| \leq \frac{C_1\eta + C_rT\eta^{n+1}}{1-\mu} + \frac{M\eta}{1-\mu} \sum_{j=1}^{k-1} \|\epsilon_j\|.$$

$$\implies \|\epsilon_k\| \leq A_\eta + B\eta \sum_{j=1}^{k-1} \|\epsilon_j\|, \qquad A_\eta := \frac{C_1\eta + C_rT\eta^{n+1}}{1-\mu}, \qquad B := \frac{M}{1-\mu}.$$

By induction and using $e^x \geq x+1$ at the end, we get:

$$\|\epsilon_k\| \leq A_\eta(1+B\eta)^{k-1} \leq A_\eta e^{B\eta k}.$$

This is clearly $O(\eta)$ since $A_\eta = O(\eta)$ and $\eta k \leq T$. $\square$

Note that if we can guarantee a better bound on ε_1, e.g. if $\|\varepsilon_1\| = O(\eta^{n+2})$, which is possible via using a piecewise structure, we can get $\sup_{0 \leq \eta \leq \frac{T}{\eta}} \|\varepsilon_n\| = O(\eta^{n+1})$ and $\sup_{0 \leq \eta \leq \frac{T}{\eta}} \|\varepsilon_n - \varepsilon_{n-1}\| = O(\eta^{n+2})$. The same argument from the claim will work, except we will have $A_\eta = O(\eta^{n+1})$.

6. Stability analysis

Let λ_i and u_i be the eigenvalues and eigenvectors of the Hessian $\nabla^2 f(x)$; then the Principal Flow is defined as:

$$\dot{x} = \sum_{i=0}^{D-1} \frac{\log(f(\mu, \eta\lambda_i))}{\eta\lambda_i} \big(\nabla f^T u_i\big) u_i.$$

Consider a critical point x^* and let λ_i^*, u_i^* be the eigenvalues and eigenvectors of the Hessian $\nabla^2 f(x^*)$; the Jacobian at x^* can be calculated, similarly to Rosca et al 2023 section A.4, as:

$$J(x^*) = \sum_{i=0}^{D-1} \frac{\log(f(\mu, \eta\lambda_i^*))}{\eta}\, u_i^* (u_i^*)^T,$$

whose eigenvalues are $\frac{\log(f(\mu,\eta\lambda_i^*))}{\eta}$. For that critical point to be exponentially asymptotically attractive, all eigenvalues of the Jacobian evaluated at x^* need to have strictly negative real part.

Case 1. x^* is a local minimum, i.e. $\forall \lambda_i^* \geq 0$.

Note that $f(\mu, x)$ is real for $x > (\sqrt{\mu}+1)^2$ or $x < (1-\sqrt{\mu})^2$, $f(\mu, x)$ is decreasing (whenever its real) with $f(\mu, 0) = 1$, $f(\mu, x)$ is positive when $x < \mu + 1$ (and whenever it's real) and $f(\mu, x) = -1$ iff $x = 2(\mu + 1)$. Since $\eta\lambda_i^* > 0$, we have $f(\mu, \eta\lambda_i^*) < 1$ (if it's real), so there are four cases:

- $1 > f(\mu, \eta\lambda_i^*) > 0$, which happens when $0 < x < (1 - \sqrt{\mu})^2$, i.e. $\boxed{\eta < \frac{(1-\sqrt{\mu})^2}{\lambda_i^*}}$. In this case the eigenvalue $\frac{\log(f(\mu,\eta\lambda_i^*))}{\eta} < 0$, so the principal vector field is **attractive** in the corresponding eigenvector direction.
- $0 \geq f(\mu, \eta\lambda_i^*) > -1$, which happens for $x \geq (\sqrt{\mu} + 1)^2$, i.e. $\boxed{\frac{2(\mu+1)}{\lambda_i^*} > \eta \geq \frac{(\sqrt{\mu}+1)^2}{\lambda_i^*}}$. In this case the real part of the eigenvalue is $\frac{\log(-f(\mu,\eta\lambda_i^*))}{\eta} < 0$, so the principal vector field is **attractive** in the corresponding eigenvector direction.
- $f(\mu, \eta\lambda_i^*) \leq -1$, which happens for $\boxed{\eta \geq \frac{2(\mu+1)}{\lambda_i^*}}$. In this case the real part of the eigenvalue is non-negative, so the principal vector field is **not attractive** in the corresponding eigenvector direction, and if $\eta > \frac{2(\mu+1)}{\lambda_i^*}$ it is **repelled** in the corresponding eigenvector direction.
- $f(\mu, \eta\lambda_i^*)$ is complex, which happens for $\boxed{\frac{(\sqrt{\mu}+1)^2}{\lambda_i^*} > \eta \geq \frac{(1-\sqrt{\mu})^2}{\lambda_i^*}}$. Then $|f(\mu, z_i)| = \sqrt{\mu} < 1$, so the real part is negative, i.e. the complexified principal flow is **attractive** and oscillatory in the corresponding eigenvector direction.

Thus, PF is not always attracted to local minima and can be repelled around local minima for large learning rates.

Case 2. x^* is a strict saddle point, so $\exists \lambda_s^* < 0$, for which $f(\mu, \eta\lambda_s^*) > 1$, so the eigenvalue $\frac{\log(f(\mu,\eta\lambda_s^*))}{\eta} > 0$, i.e. the PF is **repelled** in the eigendirections of strict saddle points.

7. Piecewise model and improved error guarantee

We use the model from Lyu et al 2025, but we are going to incorporate the principle of ignoring higher-order gradients of $f(x)$ in $I_k^{\pm}$ and not truncate the series γ_k, in the spirit of Rosca's work.

Let

$$\dot{x}(t) = -G_k(x(t)) - \eta\gamma_k(x(t)) := -V_k(x(t)), \qquad t \in [t_k, t_{k+1}), \quad t_k = k\eta.$$

Fix $k \geq 0$, shift $t_k = k\eta$ to time 0, and assume $x(t_k) = x(0)$ and

$$\dot{x} = -V_k(x), \qquad t > 0 \text{ and } \dot{x} = -V_{k-1}(x), \qquad t < 0.$$

Under the principle of ignoring higher-order derivatives of f in the Taylor expansions, we are matching

$$x(\eta) + \mu x(-\eta) \text{ with } (1+\mu)x(0) - \eta\nabla f(x(0)).$$

Therefore,

$$x(\eta) + \mu x(-\eta) = (1+\mu)x(0) + \sum_{q=1}^{\infty} \frac{\eta^q}{q!} \left[x_+^{(q)}(0) + \mu(-1)^q x_-^{(q)}(0) \right], \text{where}$$

$$x_+^{(q)}(0) := \lim_{t \downarrow 0} x^{(q)}(t), \qquad x_-^{(q)}(0) := \lim_{t \uparrow 0} x^{(q)}(t).$$

We require

$$\sum_{q=1}^{\infty} \frac{\eta^q}{q!} \left[x_+^{(q)}(0) + \mu(-1)^q x_-^{(q)}(0) \right] = -\eta\nabla f(x(0)).$$

The $q = 1$ term above is (where all functions are evaluated at $x(0)$):

$$\begin{aligned} \eta \left[x_+^{(1)}(0) - \mu x_-^{(1)}(0) \right] &= \eta \left[-G_k - \eta\gamma_k + \mu G_{k-1} + \mu\eta\gamma_{k-1} \right] \\ &= -\eta \left(G_k - \mu G_{k-1} \right) - \eta^2 \left(\gamma_k - \mu\gamma_{k-1} \right). \end{aligned}$$

To match the RHS, as in Lyu et al, we require

$$G_k - \mu G_{k-1} = \nabla f, \; G_{-1} = 0 \implies G_k = \frac{1-\mu^{k+1}}{1-\mu} \nabla f.$$

Hence, we require:

$$\gamma_k - \mu\gamma_{k-1} = \sum_{q=2}^{\infty} \frac{\eta^{q-2}}{q!} \left[x_+^{(q)}(0) + \mu(-1)^q x_-^{(q)}(0) \right].$$

Let

$$g(x) := -\frac{\nabla f(x)}{1-\mu}; \; \gamma_k = \sum_{p=0}^{\infty} \eta^p \gamma_k^{(p)}; \; \boxed{\dot{x} = \sum_{p=0}^{\infty} \eta^p g_{k,p}(x)}; \; g_{k,0} := (1-\mu^{k+1})g; \; g_{k,p} := -\gamma_k^{(p-1)}.$$

As with the non-piecewise case, we have:

$$x_+^{(q)}(0) = \sum_{m \geq 0} \eta^m \frac{d^{q-2}}{dt^{q-2}} \left(\sum_{i+j=m} \nabla_x g_{k,j} \, g_{k,i} \right) \Big|_{t=0^+}; \quad x_-^{(q)}(0) = \sum_{m \geq 0} \eta^m \frac{d^{q-2}}{dt^{q-2}} \left(\sum_{i+j=m} \nabla_x g_{k-1,j} \, g_{k-1,i} \right) \Big|_{t=0^-}.$$

Let

$$\frac{d^{q-2}}{dt^{q-2}} \left(\sum_{i+j=m} \nabla_x g_{r,j} \, g_{r,i} \right) = \sum_{p \geq 0} \eta^p A_{q,m}^p[r], \qquad q \geq 2.$$

Thus, we have:

$$x_+^{(q)}(0) = \sum_{m \geq 0} \sum_{p \geq 0} \eta^{m+p} A_{q,m}^p[k], \; x_-^{(q)}(0) = \sum_{m \geq 0} \sum_{p \geq 0} \eta^{m+p} A_{q,m}^p[k-1].$$

Note that

$$\gamma_k - \mu\gamma_{k-1} = -\sum_{n=1}^{\infty} \eta^{n-1} \left(g_{k,n} - \mu g_{k-1,n} \right) \text{ and}$$

$$\sum_{q=2}^{\infty}\frac{\eta^{q-2}}{q!}\left[x_+^{(q)}(0)+\mu(-1)^q x_-^{(q)}(0)\right]$$
$$=\sum_{q=2}^{\infty}\frac{\eta^{q-2}}{q!}\left[\sum_{m,p\geq 0}\eta^{m+p}A_{q,m}^p[k]+\mu(-1)^q\sum_{m,p\geq 0}\eta^{m+p}A_{q,m}^p[k-1]\right].$$
$$=\sum_{q=2}^{\infty}\sum_{m,p\geq 0}\frac{\eta^{q-2+m+p}}{q!}\left[A_{q,m}^p[k]+\mu(-1)^q A_{q,m}^p[k-1]\right].$$

Hence, we require

$$g_{k,n}-\mu g_{k-1,n}=-\sum_{q=2}^{n+1}\frac{1}{q!}\sum_{m=0}^{n+1-q}\left[A_{q,m}^{n+1-q-m}[k]+\mu(-1)^q A_{q,m}^{n+1-q-m}[k-1]\right].$$

We claim that:

$$\boxed{g_{k,p}=c_{k,p}(\nabla_x g)^p g+C(\nabla^2 g)}.$$

As before, we have:

$$A_{q,m}^p[r]=\sum_{i+j=m}\sum_{\substack{\ell_1+\cdots+\ell_{q-2}=p\\ \ell_s\geq 0}}c_{r,i}c_{r,j}c_{r,\ell_1}\cdots c_{r,\ell_{q-2}}(\nabla_x g)^{m+p+q-1}g+C(\nabla_x^2 g).$$

Thus,

$$c_{k,n}-\mu c_{k-1,n}=-\sum_{q=2}^{n+1}\frac{1}{q!}\sum_{\substack{\lambda_1+\cdots+\lambda_q=n+1-q\\ \lambda_s\geq 0}}\left[c_{k,\lambda_1}\cdots c_{k,\lambda_q}+\mu(-1)^q c_{k-1,\lambda_1}\cdots c_{k-1,\lambda_q}\right] \quad .$$

Since $c_{k,0}=1-\mu^{k+1}, c_{k-1,0}=1-\mu^k$, i.e. $c_{k,0}-\mu c_{k-1,0}=1-\mu$ if $c_k(x):=\sum_{n=0}^{\infty}c_{k,n}x^n$, we have:

$$c_k(x)-\mu c_{k-1}(x)=1-\mu-\frac{1}{x}\sum_{q=2}^{\infty}\frac{x^q}{q!}\left[c_k(x)^q+\mu(-1)^q c_{k-1}(x)^q\right].$$

$$c_k-\mu c_{k-1}=1-\mu-\frac{1}{x}\Big[e^{xc_k}-1-xc_k+\mu\left(e^{-xc_{k-1}}-1+xc_{k-1}\right)\Big].$$
$$e^{xc_k(x)}+\mu e^{-xc_{k-1}(x)}=(1-\mu)x+1+\mu.$$

Since $c_{-1}(x)=0$, if $y_k(x)=e^{xc_k(x)}$, $y_{-1}(x)=1$ we have

$$c_0(x)=\frac{1}{x}\log\left(1+(1-\mu)x\right)$$

$$\boxed{y_k(x)=(1-\mu)x+1+\mu-\frac{\mu}{y_{k-1}(x)},\quad c_k(x)=\frac{1}{x}\log y_k(x)}.$$

If we let $k\to\infty$, we clearly recover our non-piecewise PF.
Now consider the truncated ODE:

$$\dot{x}=\sum_{p=0}^{n}\eta^p g_{k,p}=-G_k-\eta\sum_{\sigma=0}^{n-1}\eta^\sigma\gamma_k^{(\sigma)}\text{ on }[t_k;t_{k+1}).$$

We extend the ODE to $[-\eta,0)$ with $G_{-1}=\gamma_{-1}=0$ so that $x(-\eta)=x(0)=x_0$. Therefore, from the matching itself we get

$$\varepsilon_1=x(\eta)+\mu x(-\eta)-(1+\mu)x(0)+\eta\nabla f(x(0))=O(\eta^{n+2}),$$

and now the argument from the Claim in a previous section is applicable to get $\sup_{0\leq\eta\leq\frac{T}{\eta}}\|\varepsilon_n\|=O(\eta^{n+1})$.

II. New construction of HA-SME and its extension to momentum

1. Construction of HA-SME

Let $x, y \in \mathbb{R}^d$ be arbitrary and let $u \in C_b^\infty(\mathbb{R}^d)$ be a test function. We will find $b(x), \mathcal{D}(x)$ that define SDE

$$dX_t = b(X_t)dt + \sqrt{\mathcal{D}(X_t)}dW_t$$

so that

$$\mathbb{E}(u(x_1) \mid x_0 = y) \text{ and } \mathbb{E}(u(X(\eta)) \mid X(0) = y)$$

match under the following principle (P) *for any y, while treating x as fixed* - in the expansions of u and repeated compositions of gradients, ignore all terms containing at least one of

$$\nabla^p u(y), \nabla^q f(x), \nabla^r \Sigma(x) \text{ for } p, q \geq 3, r \geq 1$$

(we will give a more precise definition below). This is a generalization of the principle used in HA-SME; requiring matching for arbitrary y is used to identify the complete affine transition and the final HA-SME equality is obtained by setting $y = x$.

Let

$$g_x := \nabla f(x), \qquad H_x := \nabla^2 f(x), \qquad \Sigma_x := \Sigma(x).$$

Following the functional structure result from HA-SME, we will use the following ansatz:

$$\boxed{b(x) = F_\eta(H_x)g_x, \mathcal{D}(x) = G_\eta(H_x, \Sigma_x)},$$

where F_η, G_η have the following series form:

$$F_\eta(H_x) = \sum_{p=0}^{\infty} c_p \eta^p H_x^p, \quad G_\eta(H_x, \Sigma_x) = \sum_{p=0}^{\infty} \Big(\sum_{k=0}^{p-1} a_{k,p-1-k} \cdot H_x^k \Sigma_x H_x^{p-1-k} \Big) \eta^p.$$

Claim. Under P for $y = x$ (the principle used in HA-SME), we have the following projections:

$$\boxed{\begin{array}{l} b(x) \to b(x), \qquad \nabla b(x) \to A(x) := F_\eta(H_x)H_x, \quad \nabla^p b(x) \to 0 \quad \text{for } p \geq 2; \\ \mathcal{D}(x) \to \mathcal{D}(x), \quad \nabla^q \mathcal{D}(x) \to 0 \quad \text{for } q \geq 1. \end{array}} \quad (\Delta)$$

Remark. This claim holds even without assuming the exact series form in the ansatz. It allows us to apply directly the principle of ignoring higher-order gradients of f, Σ to $b, \mathcal{D}$ instead of working inductively with the functions g_p, h_p from their η-series, which removes the necessity of using generating functions.

Proof. For convenience drop the x index in f, Σ, H.

Zeroth-order projection of the drift.

The drift has the series representation $b(x) = \sum_{n=0}^{\infty} c_n \eta^n H^n g$. Every monomial $H^n g$ contains only first- and second-order derivatives of f. Therefore, none of its factors is discarded under P, and hence $b(x)$ is preserved under P.

First derivative of the drift.

Let $h \in \mathbb{R}^d$. For each $n \geq 0$, the directional derivative of $H^n g$ is

$$D(H^n g)[h] = \sum_{j=0}^{n-1} H^j \big(DH[h]\big) H^{n-1-j} g + H^n Dg[h].$$

Since

$$DH[h] = \nabla^3 f(x)[h],$$

every term in the sum contains a third derivative of f and is discarded by P. On the other hand,

$$Dg[h] = Hh,$$

which contains only the Hessian and is retained. Consequently,

$$(D(H^n g)[h]) \overset{P}{=} H^{n+1}h.$$

Applying this identity coefficientwise to the drift series gives

$$\begin{aligned} Db(x)[h] &\overset{P}{=} \sum_{n=0}^{\infty} c_n \eta^n H^{n+1} h \\ &= \left(\sum_{n=0}^{\infty} c_n \eta^n H^n \right) Hh \\ &= F_\eta(H) Hh. \end{aligned}$$

Since this holds for every $h \in \mathbb{R}^d$,

$$\nabla b(x) \overset{P}{=} F_\eta(H) H =: A(x).$$

Higher derivatives of the drift.

Let $p \geq 2$. By the generalized product rule, every term in

$$D^p(H^n g)[h_1, \ldots, h_p]$$

is obtained by distributing the p derivatives among the n factors H and the final factor g. If at least one derivative acts on an H-factor, the resulting term contains a positive-order derivative of H. Since

$$D^r H = D^{r+2} f, \qquad r \geq 1,$$

that term contains a derivative of f of order at least three and is discarded. The only remaining possibility is that no derivative acts on any H-factor. In that case, all p derivatives act on g, and the resulting term contains

$$D^p g = D^{p+1} f.$$

Because $p \geq 2$, this is again a derivative of f of order at least three and is discarded. Thus,

$$D^p(H^n g) \overset{P}{=} 0, \qquad p \geq 2.$$

Applying this coefficientwise to the drift series yields

$$\nabla^p b(x) \overset{P}{=} 0, \qquad p \geq 2.$$

Zeroth-order projection of the covariance.

The covariance has the series representation

$$\mathcal{D}(x) = \sum_{n=1}^{\infty} \eta^n \sum_{k=0}^{n-1} a_{k,n-1-k}\, H^k \Sigma H^{n-1-k}.$$

Every undifferentiated monomial

$$H^k \Sigma H^{n-1-k}$$

contains only $H = \nabla^2 f(x)$ and the undifferentiated covariance $\Sigma(x)$, so it's preserved under P.

Positive-order derivatives of the covariance.

Fix $q \geq 1$. By the generalized product rule, every term in

$$D^q\big(H^k \Sigma H^{n-1-k}\big)$$

is obtained by distributing the q derivatives among the H-factors and the single Σ-factor. Because $q \geq 1$, at least one factor must be differentiated. There are two possibilities. First, if a derivative acts on an H-factor, the resulting term contains

$$D^r H = D^{r+2} f$$

for some $r \geq 1$. It therefore contains a derivative of f of order at least three and is discarded. Second, if no derivative acts on an H-factor, then at least one derivative acts on Σ. The resulting term contains

$$D^r \Sigma$$

for some $r \geq 1$, and is also discarded under P. Hence every term in the fully expanded derivative is discarded:

$$D^q\big(H^k \Sigma H^{n-1-k}\big) \overset{P}{=} 0, \qquad q \geq 1.$$

Applying this coefficientwise to the covariance series gives

$$\nabla^q \mathcal{D}(x) \overset{P}{=} 0, \qquad q \geq 1.$$

This proves the claim. □

1.1. Definition of a projection map $\Pi_{x,y}$

The goal of this section is to formally define a map $\Pi_{x,y}$ of ignoring higher-order gradients under P in the following two situations, so that it is consistent with the definition from HA-SME for $x = y$:

$$\Pi_{x,y}\big[\mathbb{E}\big(u(x_1) \mid x_0 = y\big)\big] \text{ and } \Pi_{x,y}\big[\mathbb{E}\big(u(X(\eta)) \mid X(0) = y\big)\big].$$

Below q denotes a dummy variable on which the generator acts; we assume x is fixed when q varies.
Let

$$Q_{u,y}(q) = u(y) + \nabla u(y) \cdot (q - y) + \frac{1}{2}\nabla^2 u(y) : (q - y)(q - y)^\top$$

Consider the following approximated process:

$$\widetilde{x}_1^x = \widetilde{x}_0^x - \eta\big(g_x + H_x(\widetilde{x}_0^x - x)\big) + \widetilde{\varepsilon}, \quad \widetilde{x}_0^x = y, \qquad \mathbb{E}(\widetilde{\varepsilon}) = 0, \quad \mathrm{Cov}(\widetilde{\varepsilon}) = \eta^2 \Sigma_x.$$

The motivation for this comes from considering the Taylor expansions about x of $\nabla f(y)$ and $\Sigma(y)$ under P, because both are involved in the update $x_1 = y - \eta \nabla F(y; \varepsilon)$.
Then we define

$$\boxed{\Pi_{x,y}\mathbb{E}[u(x_1) \mid x_0 = y] := \mathbb{E}\big[Q_{u,y}\big(\widetilde{x}_1^x\big) \mid \widetilde{x}_0^x = y\big].}$$

Explicitly,

$$\begin{aligned}\Pi_{x,y}\mathbb{E}[u(x_1) \mid x_0 = y] = u(y) &- \eta\big(g_x + H_x(y - x)\big) \cdot \nabla u(y) \\ &+ \frac{\eta^2}{2}\Big[\big(g_x + H_x(y - x)\big)\big(g_x + H_x(y - x)\big)^\top + \Sigma_x\Big] : \nabla^2 u(y).\end{aligned}$$

Similarly, define a continuous approximated process $\widetilde{X}^x(t)$ given by the SDE (OU process):

$$d\widetilde{X}^x(t) = (b(x) + A(x)(\widetilde{X}^x(t) - x))dt + \sqrt{\mathcal{D}(x)}dW_t, \quad \widetilde{X}^x(0) = y.$$

with the following generator:

$$\widetilde{L}_x u(q) = \big(b(x) + A(x)(q - x)\big) \cdot \nabla_q u(q) + \frac{1}{2}\mathcal{D}(x) : \nabla_q^2 u(q) := \widetilde{b}_x(q) \cdot \nabla u(q) + \frac{1}{2}\widetilde{\mathcal{D}}_x(q) : \nabla^2 u(q).$$

The motivation for this approximation comes from considering the Taylor expansions of $b(q)$, $\mathcal{D}(q)$ with respect to x and applying P to the gradients of $b, \mathcal{D}$ in the form of (Δ).
The solution of this OU process is given by:

$$\widetilde{X}^x(\eta) = x + e^{\eta A(x)}(y - x) + \int_0^\eta e^{sA(x)} b(x)\, ds + \varepsilon_{\eta,x}, \quad \mathbb{E}(\varepsilon_{\eta,x}) = 0, \quad \mathrm{Cov}(\varepsilon_{\eta,x}) = \int_0^\eta e^{sA(x)} \mathcal{D}(x) e^{sA(x)^\top}\, ds.$$

Then define:

$$\boxed{\Pi_{x,y}\big[\mathbb{E}\big(u(X(\eta)) \mid X(0) = y\big)\big] = \mathbb{E}\Big[Q_{u,y}\big(\widetilde{X}^x(\eta)\big) \mid \widetilde{X}^x(0) = y\Big].}$$

Explicitly,

$$\Pi_{x,y}\mathbb{E}[u(X(\eta)) \mid X(0) = y] := \sum_{i=0}^{\infty} \frac{\eta^i}{i!} \big[\widetilde{L}_x^i Q_{u,y}\big](y).$$

Note that for both the discrete and continuous definitions the approximated process represents ignoring higher-order gradients of $f(x), \Sigma(x)$ (by applying P in the discrete update and the SDE), while $Q_{u,y}$ represents ignoring higher-order derivatives of $u(y)$. Conceptually, this is exactly what we expect from $\Pi_{x,y}$, but in the next section we will show why this is exactly the principle used in HA-SME for $y = x$.

1.2. Consistency of the definitions with the projection from HA-SME

Now we show these definitions agree with the ones from HA-SME for $y = x$.

Clearly the discrete definition is consistent with HA-SME since the Hessian terms disappear due to $y = x$:

$$\begin{aligned}\Pi_{x,x}\mathbb{E}[u(x_1) \mid x_0 = x] = u(x) - \eta\nabla f(x)\cdot\nabla u(x) \\ + \frac{\eta^2}{2}\left(\nabla f(x)\nabla f(x)^\top + \Sigma(x)\right) : \nabla^2 u(x).\end{aligned}$$

For the continuous one, we have

$$\widetilde{L}_x\varphi(q) = \big(b(x) + A(x)(q-x)\big)\cdot\nabla_q\varphi(q) + \frac{1}{2}\mathcal{D}(x) : \nabla_q^2\varphi(q)$$

We have to show that $\forall i \geq 0$, if L is the generator of the SDE for $X(t)$,

$$\boxed{\Pi_x L^i u(x) = \widetilde{L}_x^i Q_{u,x}(x)}$$

The projection of the LHS is defined as in HA-SME (denote it by $\Pi_{x,x} := \Pi_x$): it first ignores the terms containing higher-order gradients of u in the expansions of the compositions of L and then applies (Δ) to the gradients of $b(x), \mathcal{D}(x)$, which represents the ignorance of higher-order gradients of f, Σ in the η-series of $b, \mathcal{D}$ under our ansatz.

Note that since $Q_{u,x}(q)$ is a quadratic in q, we have that:

$$\nabla_q^p Q_{u,x}(x) = \begin{cases}\nabla^p u(x), & p = 0,1,2,\\ 0, & p \geq 3.\end{cases}$$

This means that

$$\boxed{\nabla^p Q_{u,x}(x) = \Pi_x(\nabla^p u(x))\,, \qquad p \geq 0}$$

By (Δ) we have that:

$$\boxed{\nabla_q^p \widetilde{b}_x(q)|_{q=x} = \Pi_x(\nabla^p b(x))\,, \qquad \nabla_q^p \widetilde{\mathcal{D}}_x(q)|_{q=x} = \Pi_x(\nabla^p \mathcal{D}(x))\,, \qquad p \geq 0.}$$

For each $i \geq 0$, there is a differential polynomial $\mathscr{P}_i$ such that

$$L^i u = \mathscr{P}_i(J_x u, J_x b, J_x \mathcal{D})\,,$$

where $J_x\phi$ denotes the finite collection of derivatives of ϕ at x appearing in the expansion.

Likewise,

$$\widetilde{L}_x^i Q_{u,x} = \mathscr{P}_i\Big(J_x Q_{u,x}, J_x\widetilde{b}_x, J_x\widetilde{\mathcal{D}}_x\Big)\,.$$

Because Π_x acts monomialwise,

$$\Pi_x\big(L^i u\big) = \mathscr{P}_i\Big(\Pi_x J_x u,\, \Pi_x J_x b,\, \Pi_x J_x \mathcal{D}\Big).$$

But from the previous two boxed equations, the three projected entries above are:

$$J_x Q_{u,x}, \qquad J_x\widetilde{b}_x, \qquad J_x\widetilde{\mathcal{D}}_x,$$

which finishes the argument that the definitions (ours and the one from HA-SME) of Π_x for a continuous process are consistent with each other.

1.3. Core Construction

Due to the definitions in 1.1, our required matching becomes:

$$\Pi_{x,y}\mathbb{E}[u(x_1) \mid x_0 = y] = \Pi_{x,y}\big[\mathbb{E}\big(u(X(\eta)) \mid X(0) = y\big)\big] \iff$$

$$\mathbb{E}\big[Q_{u,y}\big(\widetilde{x}_1^x\big) \mid \widetilde{x}_0^x = y\big] = \mathbb{E}\Big[Q_{u,y}\big(\widetilde{X}^x(\eta)\big) \mid \widetilde{X}^x(0) = y\Big].$$

Let $G_d = GL(d) \times \mathbb{R}^d \times \mathrm{Sym}_d$. Define a map $\Pi : G_d \times C_b(\mathbb{R}^d) \to C_b(\mathbb{R}^d)$ as follows:

$$[\Pi_{Y_x} u](y) := \mathbb{E}\big[Q_{u,y}\big(S_{Y_x}(y)\big)\big], \text{where}$$

$$Y_x = (M_x, r_x, K_x) \in G_d, \quad S_{Y_x}(y) = M_x y + r_x + \varepsilon_x, \quad \mathbb{E}[\varepsilon_x] = 0, \qquad \mathrm{Cov}(\varepsilon_x) = K_x.$$

Note that

$$\widetilde{x}_1^x = (I - \eta H(x))y + \eta H(x)x - \eta g(x) + \varepsilon_x = S_{Y_{1,x}}(y), \quad \text{where}$$

$$\boxed{Y_{1,x} = \big(I - \eta H(x),\, \eta H(x)x - \eta g(x),\, \eta^2\Sigma(x)\big)} \text{ and}$$

$$\widetilde{X}^x(\eta) = e^{\eta A(x)}y + \big(I - e^{\eta A(x)}\big)x + \int_0^\eta e^{sA(x)}b(x)\,ds + \varepsilon_{\eta,x} = S_{Y_{2,x}(y)}, \text{where}$$

$$\boxed{Y_{2,x} = \left(e^{\eta A(x)},\, (I - e^{\eta A(x)})x + \int_0^\eta e^{sA(x)}b(x)\,ds,\, \int_0^\eta e^{sA(x)}\mathcal{D}(x)e^{sA(x)^\top}\,ds\right).}$$

Therefore, the matching becomes:

$$[\Pi_{Y_{1,x}} u](y) = [\Pi_{Y_{2,x}} u](y),$$

so since we want this equality to hold for any y, it's sufficient $b(x), \mathcal{D}(x)$ to satisfy:

$$\boxed{Y_{1,x} = Y_{2,x}}.$$

Drop the x index in A, H, g, Σ for convenience in the remainder of the calculation.

$$H = U\Lambda U^T.$$

$$e^{\eta A} = I - \eta H$$

$$A = \frac{1}{\eta}\log(I - \eta H)$$

$$A = U\,\mathrm{diag}(\beta_i)U^T$$

$$\beta_i = \frac{1}{\eta}\log(1 - \eta\lambda_i)$$

$$A^{-1}(e^{\eta A} - I)b = A^{-1}(-\eta H)b = -\eta g \iff b = H^{-1}Ag$$

$$\implies b = U\log(I - \eta\Lambda)(\eta\Lambda)^{-1}U^T g$$

$$\hat{\mathcal{D}} = U^T\mathcal{D}U, \quad \hat{\Sigma} = U^T\Sigma U \implies \hat{\mathcal{D}}_{ij}\int_0^\eta e^{s(\beta_i+\beta_j)}\,ds = \eta^2\hat{\Sigma}_{ij}$$

$$\implies \hat{\mathcal{D}}_{ij}\frac{e^{\eta(\beta_i+\beta_j)} - 1}{\beta_i + \beta_j} = \eta^2\hat{\Sigma}_{ij}$$

$$\hat{\mathcal{D}}_{ij} = \frac{\eta^2(\beta_i+\beta_j)}{e^{\eta(\beta_i+\beta_j)}-1}\hat{\Sigma}_{ij}.$$

Therefore,

$$\mathcal{D} = USU^T,$$

where

$$S_{ij} = \frac{\log((1-\eta\lambda_i)(1-\eta\lambda_j))}{\eta\lambda_i\lambda_j - \lambda_i - \lambda_j}\hat{\Sigma}_{ij}.$$

Note that the expressions for $b, \mathcal{D}$ satisfy our ansatz and also agree with what was established in HA-SME.

2. Extension to SGD with momentum

In this section we work with triples $(\mathbf{M}, \mathbf{r}, \mathbf{K}) \in G_{2d} = GL(2d) \times \mathbb{R}^{2d} \times \mathrm{Sym}_{2d}$ and apply the definitions from 1.1 in phase space. Define $\Psi(q^+, q^-) = u(\frac{q^+ + \mu q^-}{\mu+1})$, or

$$\Psi(\mathbf{q}) = u(C_\mu \mathbf{q}), \qquad C_\mu = \frac{1}{1+\mu}\begin{pmatrix} I & \mu I \end{pmatrix}$$

Define the map

$$\Pi : G_{2d} \times \{\Psi : \Psi = u \circ C_\mu, u \in C_b^\infty(\mathbb{R}^d)\} \longrightarrow C_b(\mathbb{R}^d),$$

given by:

$$[\Pi_Y \Psi](\mathbf{y}) := \mathbb{E}\big[Q_{\Psi,\mathbf{y}}\big(S_{Y_x}(\mathbf{y})\big)\big].$$

Note that

$$\mathcal{R}_\mu(Y_{1,x}) = \mathcal{R}_\mu(Y_{2,x}) \implies [\Pi_{Y_{1,x}}\Psi](\mathbf{y}) = [\Pi_{Y_{2,x}}\Psi](\mathbf{y}), \text{where}$$

$$\mathcal{R}_\mu(\mathbf{M}_x, \mathbf{r}_x, \mathbf{K}_x) = \big(C_\mu \mathbf{M}_x J,\ C_\mu \mathbf{r}_x,\ C_\mu \mathbf{K}_x C_\mu^\top\big), \qquad J = \begin{pmatrix} I \\ I \end{pmatrix}$$

We first show how the previous results extend for the discrete process.

$$x_{k+1} = x_k - \eta\nabla F(x_k;\varepsilon_k) + \mu(x_k - x_{k-1});\ x_0 = x_{-1} = y;$$

Let

$$\mathbf{x}_1 := \begin{pmatrix} x_1 \\ x_{-1} \end{pmatrix}, \qquad \mathbf{x}_0 = \mathbf{y} := \begin{pmatrix} y \\ y \end{pmatrix}$$

Then we have

$$\mathbf{x}_1 = \begin{pmatrix} (1+\mu)I & -\mu I \\ 0 & I \end{pmatrix}\mathbf{x} + \begin{pmatrix} -\eta\nabla f(y) \\ 0 \end{pmatrix} + \varepsilon, \mathbb{E}[\varepsilon] = 0, \quad \mathrm{Cov}(\varepsilon) = \begin{pmatrix} \eta^2\Sigma(y) & 0 \\ 0 & 0 \end{pmatrix}$$

$$\widetilde{\mathbf{x}}_1^x = \begin{pmatrix} (1+\mu)I & -\mu I \\ 0 & I \end{pmatrix}\mathbf{y} + \begin{pmatrix} -\eta\big(g_x + H_x(y-x)\big) \\ 0 \end{pmatrix} + \widetilde{\boldsymbol{\varepsilon}}_x, \quad \mathbb{E}[\widetilde{\boldsymbol{\varepsilon}}_x] = 0, \qquad \mathrm{Cov}(\widetilde{\boldsymbol{\varepsilon}}_x) = \begin{pmatrix} \eta^2\Sigma_x & 0 \\ 0 & 0 \end{pmatrix},$$

where

$$\widetilde{x}_0^x = \widetilde{x}_{-1}^x = y.$$

$$\boxed{\Pi_{x,y}[\mathbb{E}(\Psi(\mathbf{x_1}) \mid \mathbf{x}_0 = \mathbf{y})] = \mathbb{E}\left[Q_{\Psi,\mathbf{y}}\big(\widetilde{\mathbf{x}}_1^x\big) \mid \widetilde{\mathbf{x}}_0^x = \mathbf{y}\right] = [\Pi_{Y_{1,x}}\Psi](\mathbf{y})},$$

where

$$\boxed{Y_{1,x} = \left(\begin{pmatrix} (1+\mu)I - \eta H_x & -\mu I \\ 0 & I \end{pmatrix}, \begin{pmatrix} \eta H_x x - \eta g_x \\ 0 \end{pmatrix}, \begin{pmatrix} \eta^2\Sigma_x & 0 \\ 0 & 0 \end{pmatrix}\right).}$$

$$\boxed{\mathcal{R}_\mu(Y_{1,x}) = \left(I - \frac{\eta}{1+\mu}H_x,\ \frac{\eta}{1+\mu}(H_x x - g_x),\ \frac{\eta^2}{(1+\mu)^2}\Sigma_x\right).}$$

For the continuous one, instead of the SDE for $X(t)$, inspired by the GD case, we will consider two "forward" and "backward" processes $X^+(t), X^-(t)$, given by the following SDEs, where the Brownian motions $W_t^\pm$ are independent (the drift of the backward one is the negated drift of the forward one, but the diffusion is not negated):

$$dX_t^+ = b(X_t^+)\,dt + \sqrt{\mathcal{D}(X_t^+)}\,dW_t^+,$$
$$dX_t^- = -b(X_t^-)\,dt + \sqrt{\mathcal{D}(X_t^-)}\,dW_t^-.$$

Let $\boldsymbol{X}_t := \begin{pmatrix} X_t^+ \\ X_t^- \end{pmatrix} \in \mathbb{R}^{2d}$ and $\boldsymbol{W}_t := \begin{pmatrix} W_t^+ \\ W_t^- \end{pmatrix}$, so the SDE for $\boldsymbol{X}_t$ is

$$d\boldsymbol{X}_t = \boldsymbol{b}(\boldsymbol{X}_t)dt + \sqrt{\boldsymbol{\mathcal{D}}(\boldsymbol{X}_t)}d\boldsymbol{W}_t, \qquad \boldsymbol{b}(x_+, x_-) = \begin{pmatrix} b(x_+) \\ -b(x_-) \end{pmatrix} \text{ and } \boldsymbol{\mathcal{D}}(x_+, x_-) = \begin{pmatrix} \mathcal{D}(x_+) & 0 \\ 0 & \mathcal{D}(x_-) \end{pmatrix}$$

We will use the SDE for the forward process (with the same ansatz for $b, \mathcal{D}$ as before) to model SGD with momentum. In this section we will use the following matching:

$$\Pi_{x,y}(\mathbb{E}(u(\frac{x_1 + \mu x_{-1}}{\mu+1}) \mid x_0 = x_{-1} = y)) = \Pi_{x,y}(\mathbb{E}(u(\frac{X^+(\eta) + \mu X^-(\eta)}{\mu+1}) \mid \boldsymbol{X}(0) = \boldsymbol{y})) \iff$$

$$\boxed{\Pi_{x,y}[\mathbb{E}(\Psi(\mathbf{x_1}) \mid \mathbf{x}_0 = \mathbf{y})] = \Pi_{x,y}[\mathbb{E}(\Psi(\mathbf{X}(\eta)) \mid \mathbf{X}(0) = \mathbf{y})]}$$

The continuous approximated process is given by:

$$d\widetilde{\mathbf{X}}_t^x = \left(\mathbf{A}_x \widetilde{\mathbf{X}}_t^x + \mathbf{a}_x\right)dt + \sqrt{\mathcal{D}_x}\,d\mathbf{W}_t, \quad \mathbf{A}_x := \begin{pmatrix} A_x & 0 \\ 0 & -A_x \end{pmatrix}, \qquad \mathbf{a}_x := \begin{pmatrix} b_x - A_x x \\ -b_x + A_x x \end{pmatrix}, \quad \mathcal{D}_x := \begin{pmatrix} \mathcal{D}_x & 0 \\ 0 & \mathcal{D}_x \end{pmatrix}$$

with generator (the $\pm$ in the gradients denotes derivative wrt $q_\pm$):

$$\begin{aligned} \widetilde{\mathbf{L}}_x \Phi(q^+, q^-) = \; & \left[b_x + A_x(q^+ - x)\right] \cdot \nabla_+ \Phi(q^+, q^-) \\ & - \left[b_x + A_x(q^- - x)\right] \cdot \nabla_- \Phi(q^+, q^-) \\ & + \frac{1}{2}\mathcal{D}_x : \nabla_{++}^2 \Phi(q^+, q^-) + \frac{1}{2}\mathcal{D}_x : \nabla_{--}^2 \Phi(q^+, q^-). \end{aligned}$$

As with the discrete case, we have the following chain:

$$\boxed{\Pi_{x,y}\left[\mathbb{E}\big(\Psi\big(\mathbf{X}(\eta)\big) \,\big|\, \mathbf{X}(0) = \mathbf{y}\big)\right] = \mathbb{E}\Big[Q_{\Psi,\mathbf{y}}\big(\widetilde{\mathbf{X}}^x(\eta)\big) \,\Big|\, \widetilde{\mathbf{X}}^x(0) = \mathbf{y}\Big] = [\Pi_{Y_{2,x}}\Psi](\mathbf{y})}, \text{ where}$$

$$\boxed{Y_{2,x} = \left(\begin{pmatrix} e^{\eta A_x} & 0 \\ 0 & e^{-\eta A_x} \end{pmatrix}, \begin{pmatrix} (I - e^{\eta A_x})x + \displaystyle\int_0^\eta e^{sA_x} b_x\,ds \\ (I - e^{-\eta A_x})x - \displaystyle\int_0^\eta e^{-sA_x} b_x\,ds \end{pmatrix}, \begin{pmatrix} \displaystyle\int_0^\eta e^{sA_x}\mathcal{D}_x e^{sA_x^\top}\,ds & 0 \\ 0 & \displaystyle\int_0^\eta e^{-sA_x}\mathcal{D}_x e^{-sA_x^\top}\,ds \end{pmatrix} \right).}$$

$$\boxed{\begin{aligned} \mathcal{R}_\mu(Y_{2,x}) = \Bigg(& \frac{e^{\eta A_x} + \mu e^{-\eta A_x}}{1+\mu}, \\ & \left(I - \frac{e^{\eta A_x} + \mu e^{-\eta A_x}}{1+\mu}\right) x \\ & \quad + \frac{\displaystyle\int_0^\eta e^{sA_x} b_x\,ds - \mu \int_0^\eta e^{-sA_x} b_x\,ds}{1+\mu}, \\ & \frac{1}{(1+\mu)^2}\int_0^\eta \left(e^{sA_x}\mathcal{D}_x e^{sA_x^\top} + \mu^2 e^{-sA_x}\mathcal{D}_x e^{-sA_x^\top}\right) ds \Bigg). \end{aligned}}$$

Therefore, we are left to match $\mathcal{R}_\mu(Y_{1,x}) = \mathcal{R}_\mu(Y_{2,x})$:

$$\begin{aligned} I - \frac{\eta}{\mu+1}H &= \frac{e^{\eta A} + \mu e^{-\eta A}}{\mu+1} \\ -\frac{\eta}{\mu+1}g &= A^{-1}\left(\frac{e^{\eta A} - I + \mu(e^{-\eta A} - I)}{\mu+1}\right) b \\ \frac{\eta^2}{(\mu+1)^2}\Sigma &= \int_0^\eta \frac{e^{sA}\mathcal{D}e^{sA^T} + \mu^2 e^{-sA}\mathcal{D}e^{-sA^T}}{(\mu+1)^2}\, ds \end{aligned}$$

Drop the x index in A, H, g, Σ for convenience in the remainder of the calculation.

$$H = U\Lambda U^T.$$

$$e^{\eta A} + \mu e^{-\eta A} = (\mu+1)I - \eta H \iff e^{2\eta A} - \big((\mu+1)I - \eta H\big)e^{\eta A} + \mu I = 0$$

$$A = \frac{1}{\eta}\log\left(\frac{(\mu+1)I - \eta H + \sqrt{\big((\mu+1)I - \eta H\big)^2 - 4\mu I}}{2}\right)$$

$$A = U\,\mathrm{diag}(\beta_i)U^T$$

$$\beta_i = \frac{1}{\eta}\log\left(\frac{\mu+1-\eta\lambda_i + \sqrt{(\mu+1-\eta\lambda_i)^2 - 4\mu}}{2}\right)$$

$$A^{-1}(e^{\eta A} - I + \mu(e^{-\eta A} - I))b = A^{-1}(-\eta H)b = -\eta g \iff b = H^{-1}Ag$$

$$\implies b = U\log\left(\frac{(\mu+1)I - \eta\Lambda + \sqrt{(\mu-1)^2 I - 2(\mu+1)\eta\Lambda + \eta^2\Lambda^2}}{2}\right)(\eta\Lambda)^{-1}U^T g$$

$$\hat{\mathcal{D}} = U^T\mathcal{D}U, \qquad \hat{\Sigma} = U^T\Sigma U \implies \hat{\mathcal{D}}_{ij}\int_0^\eta \left(e^{s(\beta_i+\beta_j)} + \mu^2 e^{-s(\beta_i+\beta_j)}\right) ds = \eta^2\hat{\Sigma}_{ij}$$

$$\implies \hat{\mathcal{D}}_{ij}\frac{e^{\eta(\beta_i+\beta_j)} - \mu^2 e^{-\eta(\beta_i+\beta_j)} + \mu^2 - 1}{\beta_i+\beta_j} = \eta^2\hat{\Sigma}_{ij}$$

$$\hat{\mathcal{D}}_{ij} = \frac{\eta^2(\beta_i+\beta_j)}{e^{\eta(\beta_i+\beta_j)} - \mu^2 e^{-\eta(\beta_i+\beta_j)} + \mu^2 - 1}\hat{\Sigma}_{ij}.$$

Therefore, the drift is

$$\boxed{b(x) = \frac{U(x)\,\log\big(f(\mu I, \eta\Lambda(x))\big)\,U(x)^T}{\eta\Lambda(x)}\nabla f(x)}$$

and the diffusion is

$$D(x) = \sqrt{\mathcal{D}(x)},$$

where

$$\mathcal{D}(x) = U(x)S(x)U(x)^T,$$

with

$$\boxed{S_{ij} = [U^T\Sigma U]_{ij}\frac{\eta\,\log\big(f(\mu,\eta\lambda_i)f(\mu,\eta\lambda_j)\big)}{f(\mu,\eta\lambda_i)f(\mu,\eta\lambda_j) - \dfrac{\mu^2}{f(\mu,\eta\lambda_i)f(\mu,\eta\lambda_j)} + \mu^2 - 1}}$$

where

$$f(\mu,x)=\frac{\mu+1-x+\sqrt{(\mu+1-x)^2-4\mu}}{2}.$$

As we showed for GD, the series $\frac{\log(f(\mu,x))}{x}$ has radius of convergence $(1-\sqrt{\mu})^2$ and moreover

$$\boxed{\frac{\log f(\mu,x)}{x}=-\frac{1}{1-\mu}-\frac{1+\mu}{2(1-\mu)^3}x+O(x^2).}$$

Define $z(x,y)=f(\mu,x)f(\mu,y)$, and let

$$g_\mu(x,y)=\frac{\log z(x,y)}{z(x,y)-\mu^2/z(x,y)+\mu^2-1}=\frac{z(x,y)\log z(x,y)}{(z(x,y)-1)(z(x,y)+\mu^2)}.$$

The apparent singularity at $z=1$ is removable, since $\lim_{z\to1}\frac{\log z}{z-1}=1$. Consequently,

$$g_\mu(0,0)=\lim_{z\to1}\frac{\log z}{z-1}\frac{z}{z+\mu^2}=\boxed{\frac{1}{1+\mu^2}}.$$

The additional genuine pole of the corrected kernel is at $z=-\mu^2$. The branch of $f(\mu,x)$ satisfying $f(\mu,0)=1$ obeys

$$f(\mu,x)+\frac{\mu}{f(\mu,x)}=1+\mu-x.$$

If $|f(\mu,x)|=\sqrt{\mu}$, write $f(\mu,x)=\sqrt{\mu}\,e^{i\theta}$. Then $x=1+\mu-2\sqrt{\mu}\cos\theta$, and hence $x\in\left[(1-\sqrt{\mu})^2,(1+\sqrt{\mu})^2\right]$. It follows that, whenever $|x|<(1-\sqrt{\mu})^2$, $|f(\mu,x)|>\sqrt{\mu}$. Therefore, for $|x|,|y|<(1-\sqrt{\mu})^2$,

$$|z(x,y)|=|f(\mu,x)f(\mu,y)|>\mu>\mu^2,$$

so $z(x,y)\neq-\mu^2$.

Therefore the radius of convergence is

$$\boxed{|x|<(1-\sqrt{\mu})^2,\qquad |y|<(1-\sqrt{\mu})^2}$$

with leading coefficient in the series expansion

$$\boxed{g_\mu(0,0)=\frac{1}{1+\mu^2}.}$$

Therefore, both $b(x),\mathcal{D}(x)$ exist when $\boxed{\eta<\frac{(1-\sqrt{\mu})^2}{\|\nabla^2 f(x)\|}}$.

3. Boundedness of the drift and diffusion of the SDE for momentum

Here we will closely follow the proofs of Lemma 15, 16 from the HA-SME paper - the main changes are that the coefficients of the generating functions and their radii of convergence are different, but the functional structure of $b,\mathcal{D}$ is the same, so the proofs are very similar.

Lemma. Fix $0\le\mu<1$. For $n\ge0$, assume that $f\in C_b^{n+2}(\mathbb{R}^d)$, and denote

$$\lambda:=\sup_{x\in\mathbb{R}^d}\|\nabla^2 f(x)\|,\qquad s:=\sup_{x\in\mathbb{R}^d}\|\nabla f(x)\|.$$

There exist constants $\eta_0,C>0$, independent of λ and s, such that, for every η satisfying

$$0<\eta<\eta_0,\qquad \eta\lambda<C,$$

the drift b of the forward momentum process X_t^+ satisfies

$$\max_{1\le i\le d} |[b(x)]_i| \le O(s) \text{ and } \max_{1\le i\le d} \|[b]_i\|_{C^n} \le O(s+\lambda).$$

Proof. Define

$$c_\mu(z) := \begin{cases} \dfrac{\log f(\mu, z)}{z}, & z \neq 0, \\ -\dfrac{1}{1-\mu}, & z = 0. \end{cases}$$

As we showed before, the singularity at $z = 0$ is removable, and c_μ is analytic for $|z| < (1 - \sqrt{\mu})^2$. Let $c_\mu(z) = \sum_{p=0}^{\infty} c_p^{(\mu)} z^p$ be its Taylor expansion at zero. The drift can then be written as

$$b(x) = \sum_{p=0}^{\infty} c_p^{(\mu)} \eta^p \big(\nabla^2 f(x)\big)^p \nabla f(x).$$

For a matrix $A \in \mathbb{R}^{d\times d}$, the (i, j)-th entry of A^p can be expressed as

$$[A^p]_{ij} = \sum_{s_1,\dots,s_{p-1}=1}^{d} [A]_{i,s_1}[A]_{s_1,s_2}\cdots[A]_{s_{p-1},j}.$$

Consequently,

$$[b(x)]_i = \sum_{p=0}^{\infty} c_p^{(\mu)} \eta^p \sum_{s_1,\dots,s_p=1}^{d} \partial_i\partial_{s_1} f(x)\partial_{s_1}\partial_{s_2} f(x)\cdots\partial_{s_{p-1}}\partial_{s_p} f(x)\partial_{s_p} f(x).$$

The term associated with $c_p^{(\mu)}$ contains d^p summands. Each summand has p factors consisting of second-order derivatives of f and one factor consisting of a first-order derivative of f. It follows that

$$|[b(x)]_i| \le s \sum_{p=0}^{\infty} |c_p^{(\mu)}| (\eta d\lambda)^p.$$

The power series $\sum_{p=0}^{\infty} |c_p^{(\mu)}| z^p$ has the same radius of convergence as the Taylor series of c_μ, namely $(1-\sqrt{\mu})^2$. Thus, if C is chosen so that $dC < (1 - \sqrt{\mu})^2$, then $\max_{1\le i\le d} |[b(x)]_i| \le O(s)$.

We next consider derivatives of b. Let $B := \max_{3\le|\alpha|\le n+2} \|D^\alpha f\|_{L^\infty}$.

Consider first one derivative. Applying the product rule to a summand associated with $c_p^{(\mu)}$ produces two types of terms:

1. p terms containing one third-order derivative of f, $p-1$ second-order derivatives of f, and one first-order derivative of f;
2. one term containing $p + 1$ second-order derivatives of f.

Therefore,

$$\begin{aligned} |\partial_j[b(x)]_i| &\le \sum_{p=0}^{\infty} |c_p^{(\mu)}| \eta^p d^p \left(pBs\lambda^{p-1} + \lambda^{p+1}\right) \\ &= sB\eta \sum_{p=1}^{\infty} p|c_p^{(\mu)}| d^p (\eta\lambda)^{p-1} + \lambda \sum_{p=0}^{\infty} |c_p^{(\mu)}| (d\eta\lambda)^p. \end{aligned}$$

Multiplying the coefficients of a power series by p does not alter its radius of convergence, since

$$\lim_{p\to\infty} p^{1/p} = 1.$$

Hence both series are bounded whenever $\eta\lambda$ and ηB are sufficiently small. By choosing η_0 such that $\eta_0 B$ is sufficiently small, we obtain

$$|\partial_j [b(x)]_i| \le O(s+\lambda).$$

The argument is analogous for derivatives of higher order. For a multi-index α with $1 \le |\alpha| \le n$, repeated use of the product rule transforms each coordinate summand associated with $c_p^{(\mu)}$ into finitely many terms of the form

$$O\big((p+1)^{|\alpha|}\big) \left(\prod_{\ell=1}^{m} \partial^{q_\ell} f \right) (\partial^2 f)^r (\partial f)^\varepsilon,$$

where

$$q_\ell \ge 3, \qquad m + r + \varepsilon = p+1, \qquad 0 \le r \le p, \qquad 0 \le \varepsilon \le 1.$$

The number of possible forms depends only on n and d, not on p, λ, or s.

The corresponding bounds involve power series whose coefficients are $|c_p^{(\mu)}|$ multiplied by a polynomial in p of degree at most n. Such multiplication does not change the radius of convergence, because

$$\lim_{p\to\infty} (p+1)^{n/p} = 1.$$

Thus, after choosing $C > 0$ and $\eta_0 > 0$ sufficiently small so that both $d\eta\lambda$ and $d\eta B$ lie strictly within the convergence radius, all derivatives of order at most n are bounded by $O(s+\lambda)$. Therefore,

$$\max_{1\le i\le d} \|[b]_i\|_{C^n} \le O(s+\lambda).$$

□

Lemma. Fix $0 \le \mu < 1$. For $n \ge 0$, assume that $f \in C_b^{n+2}(\mathbb{R}^d)$ and

$$[\Sigma]_{ij} \in C_b^n(\mathbb{R}^d) \qquad \text{for every } 1 \le i,j \le d.$$

Set $\lambda := \sup_{x\in\mathbb{R}^d} \|\nabla^2 f(x)\|$. There exist constants $\eta_0, C > 0$, independent of λ, such that, for every η satisfying

$$0 < \eta < \eta_0, \qquad \eta\lambda < C,$$

the covariance matrix $\mathcal{D}$ of the forward momentum process X_t^+ satisfies

$$\frac{1}{\eta} \max_{1\le i,j\le d} \|[\mathcal{D}]_{ij}\|_{C^n} \le O(1).$$

Proof. Recall that

$$g_\mu(x,y) = \frac{\log\big(f(\mu,x)f(\mu,y)\big)}{f(\mu,x)f(\mu,y) - \dfrac{\mu^2}{f(\mu,x)f(\mu,y)} + \mu^2 - 1},$$

where the removable singularity at $f(\mu,x)f(\mu,y) = 1$ is filled by the value $\frac{1}{1+\mu^2}$. Previously we had shown in II.2 that $g_\mu(x,y)$ converges absolutely whenever $|x|, |y| < (1-\sqrt{\mu})^2 := R_\mu$.

Writing $H(x) := \nabla^2 f(x)$, the covariance matrix of X_t^+ has the expansion

$$\mathcal{D}(x) = \eta \sum_{p,q\ge 0} a_{p,q}^{(\mu)} \eta^{p+q} H(x)^p \Sigma(x) H(x)^q.$$

For matrices $A, B \in \mathbb{R}^{d\times d}$, each entry of $A^p B A^q$ is a sum of at most d^{p+q} terms. Each such term contains $p+q$ entries of A and one entry of B. Let $S := \max_{1\le i,j\le d} \|[\Sigma]_{ij}\|_{C^n}$. Since every entry of $H(x)$ is bounded in absolute value by λ, it follows that

$$|[\mathcal{D}(x)]_{ij}| \le \eta S \sum_{p,q\ge 0} |a_{p,q}^{(\mu)}| (\eta d\lambda)^p (\eta d\lambda)^q.$$

Choose $C > 0$ such that $dC < R_\mu$. Then $\eta\lambda < C$ implies that the preceding series is uniformly bounded. Consequently,

$$\frac{1}{\eta} \max_{1\le i,j\le d} \|[\mathcal{D}]_{ij}\|_{C^0} \le O(1).$$

We now consider derivatives. Let $B := \max_{3\le|\alpha|\le n+2} \|D^\alpha f\|_{L^\infty}$. Applying one derivative to a term $H^p \Sigma H^q$ produces terms of the following types:

$$(p+q)\,(\partial^3 f)(\partial^2 f)^{p+q-1}\Sigma$$

and

$$(\partial^2 f)^{p+q}\partial\Sigma.$$

Thus, after expanding the matrix products into coordinates,

$$\begin{aligned}|\partial_k[\mathcal{D}(x)]_{ij}| \le \eta S \sum_{p,q\ge 0} |a_{p,q}^{(\mu)}| d^{p+q}(p+q)B\eta(\eta\lambda)^{p+q-1} \\ + \eta S \sum_{p,q\ge 0} |a_{p,q}^{(\mu)}|(d\eta\lambda)^{p+q}.\end{aligned}$$

The second series is bounded as above. The first is controlled by a series of the form

$$\sum_{p,q\ge 0} (p+q)|a_{p,q}^{(\mu)}| z^p w^q.$$

Multiplying the coefficients by $p+q$ does not alter the domain of convergence, because

$$\lim_{p+q\to\infty} (p+q)^{1/(p+q)} = 1.$$

Choosing η_0 sufficiently small so that $d\eta_0 B < R_\mu$ therefore gives

$$\frac{1}{\eta} \max_{1\le i,j\le d} \|\partial_k[\mathcal{D}]_{ij}\|_{C^0} \le O(1).$$

The same reasoning applies to derivatives of arbitrary order $1 \le |\alpha| \le n$. Repeated applications of the product rule produce finitely many terms of the form

$$O\big((p+q+1)^{|\alpha|}\big) \left(\prod_{\ell=1}^{m} \partial^{r_\ell} f\right) (\partial^2 f)^r \partial^\beta \Sigma,$$

where

$$r_\ell \ge 3, \qquad m + r = p + q, \qquad |\beta| \le |\alpha|.$$

The number of such forms depends only on n and d. The resulting majorant series have coefficients of the form

$$(p+q+1)^n |a_{p,q}^{(\mu)}|.$$

These series have the same convergence domain as g_μ, since

$$\lim_{p+q\to\infty} (p+q+1)^{n/(p+q)} = 1.$$

Therefore, after choosing C and η_0 sufficiently small, all derivatives of $\mathcal{D}/\eta$ of order at most n are uniformly bounded. Hence

$$\frac{1}{\eta} \max_{1\le i,j\le d} \|[\mathcal{D}]_{ij}\|_{C^n} \le O(1).$$

□

4. Well-posedness of the SDE for SGD with momentum

Now, we will study when the model is well-defined, i.e. when $\mathcal{D}(x)$ is positive definite and when the SDE has unique strong solution. Let $f_i(x) := f\big(\mu, \eta\lambda_i(x)\big)$ and $R(\mu) = (1-\sqrt{\mu})^2$. We first establish an estimate similar to Lemma 12 in HA-SME.

Lemma 1. For $0 \le a < R(\mu)$, define

$$z_-(a) := f(\mu, a)^2, \qquad z_+(a) := f(\mu, -a)^2.$$

For $z > \mu$, define

$$h_\mu(z) := (1+\mu^2)\frac{\log z}{z - \dfrac{\mu^2}{z} + \mu^2 - 1},$$

with the removable singularity at $z = 1$ given by $h_\mu(1) := 1$. Finally, let

$$\delta_\mu(a) := \max\left\{h_\mu\big(z_-(a)\big) - 1,\, 1 - h_\mu\big(z_+(a)\big)\right\}.$$

Then, for every $x, y \in [-a, a]$,

$$\boxed{\left|(1+\mu^2)\frac{\log\big(f(\mu,x)f(\mu,y)\big)}{f(\mu,x)f(\mu,y) - \dfrac{\mu^2}{f(\mu,x)f(\mu,y)} + \mu^2 - 1} - 1\right| \le \delta_\mu(a).}$$

Proof. On $x < R(\mu)$, since $f(\mu, x)$ is the larger root of a quadratic whose roots have product μ, $f(\mu, x) > \sqrt{\mu}$, so

$$\partial_x f(\mu, x) = -\frac{f(\mu,x)^2}{f(\mu,x)^2 - \mu} < 0.$$

Thus $f(\mu, \cdot)$ is strictly decreasing and positive. For $x, y \in [-a, a]$, it follows that

$$f(\mu, a)^2 \le f(\mu, x)f(\mu, y) \le f(\mu, -a)^2.$$

Clearly $z := f(\mu, x)f(\mu, y) > \mu$. Note that

$$h_\mu(z) = \frac{\log z}{z-1}\frac{(1+\mu^2)z}{z+\mu^2}.$$

A direct differentiation gives

$$h'_\mu(z) = -(1+\mu^2)\frac{\Phi_\mu(z)}{(z-1)^2(z+\mu^2)^2},$$

where

$$\Phi_\mu(z) := (z^2+\mu^2)\log z - (z-1)(z+\mu^2).$$

We show that $\Phi_\mu(z) \ge 0$ for $z > \mu$. First, if $\mu < z < 1$, then $\mu^2 < z^2$ and $\log z - (z-1) \le 0$, so

$$\begin{aligned}\Phi_\mu(z) &= z^2\log z - z(z-1) + \mu^2\big(\log z - (z-1)\big)\\ &\ge z^2\log z - z(z-1) + z^2\big(\log z - (z-1)\big)\\ &= z\big(2z\log z + 1 - z^2\big) \ge 0.\end{aligned}$$

Indeed, $p(z) := 2z\log z + 1 - z^2$ satisfies

$$p'(z) = 2(\log z + 1 - z) \le 0, \qquad p(1) = 0.$$

If $z > 1$, then $\mu^2 < 1$ and $\log z - (z-1) \le 0$, so

$$\begin{aligned}\Phi_\mu(z) &\ge z^2\log z - z(z-1) + \log z - (z-1)\\ &= (z^2+1)\log z - (z^2-1) \ge 0.\end{aligned}$$

Indeed, $q(z) := (z^2+1)\log z - (z^2-1)$ satisfies

$$q'(z) = 2z\log z - z + \frac{1}{z}, \qquad q''(z) = 2\log z + 1 - \frac{1}{z^2} \geq 0,$$

and $q'(1) = q(1) = 0$. Hence h_μ is decreasing on (μ, ∞).

Since $z_-(a) \leq z \leq z_+(a)$, we obtain $h_\mu\big(z_+(a)\big) \leq h_\mu(z) \leq h_\mu\big(z_-(a)\big)$. Moreover, since $f(\mu, x)$ is strictly decreasing and $f(\mu, 0) = 1$, $f(\mu, a) < 1 < f(\mu, -a)$, as well as $h_\mu(1) = 1$, we have

$$z_-(a) < 1 < z_+(a), \quad h_\mu\big(z_+(a)\big) \leq 1 \leq h_\mu\big(z_-(a)\big).$$

Thus,

$$|h_\mu(z) - 1| \leq \max\left\{h_\mu\big(z_-(a)\big) - 1,\, 1 - h_\mu\big(z_+(a)\big)\right\} = \delta_\mu(a).$$

This proves the desired inequality. □

We will now use this estimate to establish sufficient conditions for $\mathcal{D}$ to be positive semi-definite, similarly to Lemma 3 in HA-SME.

Lemma 2. Fix $x \in \mathbb{R}^d$, and suppose $\eta\|H(x)\| < R(\mu) = (1-\sqrt{\mu})^2$. Then $\mathcal{D}(x)$ is positive semidefinite if either of the following conditions holds: 1. $H(x)$ and $\Sigma(x)$ commute; 2. $\Sigma(x)$ is positive definite and

$$\boxed{\delta_\mu(\eta\|H(x)\|) \leq \frac{\lambda_{\min}(\Sigma(x))}{\sqrt{d}\,\lambda_{\max}(\Sigma(x))}.}$$

Equivalently, define

$$a_{\mu,\Sigma(x)} := \sup\left\{0 \leq a < R(\mu) : \delta_\mu(a) \leq \frac{\lambda_{\min}(\Sigma(x))}{\sqrt{d}\,\lambda_{\max}(\Sigma(x))}\right\}.$$

Then condition 2 holds whenever

$$\eta\|H(x)\| \leq a_{\mu,\Sigma(x)}.$$

Proof. Write $H(x) = U\Lambda U^\top, \quad \widehat{\Sigma} = U^\top\Sigma(x)U$.

First condition

Suppose $H(x)$ and $\Sigma(x)$ commute. Since both matrices are real symmetric, they can be simultaneously diagonalized. Thus we may choose U such that $U^\top\Sigma U = \widehat{\Sigma} = \operatorname{diag}(\sigma_1, \ldots, \sigma_d), \quad \sigma_i \geq 0$. It follows that

$$\mathcal{D}(x) = \eta U \operatorname{diag}\left(\sigma_i \frac{\log(f_i^2)}{f_i^2 - \dfrac{\mu^2}{f_i^2} + \mu^2 - 1}\right)_{i=1,\ldots,d} U^\top.$$

It remains to show that

$$\frac{\log(f_i^2)}{f_i^2 - \dfrac{\mu^2}{f_i^2} + \mu^2 - 1} > 0.$$

Let $z = f_i^2$. Note that $z - \frac{\mu^2}{z} + \mu^2 - 1 = \frac{(z-1)(z+\mu^2)}{z}$. Because $z + \mu^2 > 0$, the denominator has the same sign as $z - 1$, which is also the sign of $\log z$. Therefore,

$$\frac{\log z}{z - \dfrac{\mu^2}{z} + \mu^2 - 1} > 0,$$

with continuous value $1/(1+\mu^2)$ at $z = 1$. Every diagonal entry is therefore nonnegative, proving $\mathcal{D}(x) \succeq 0$. In this case, the principal square root has the explicit form

$$D(x) = U \operatorname{diag}\left(\sqrt{\eta\sigma_i \frac{\log(f_i^2)}{f_i^2 - \dfrac{\mu^2}{f_i^2} + \mu^2 - 1}}\right)_{i=1,\ldots,d} U^\top.$$

Second condition

Define the matrix K by

$$K_{ij} := (1+\mu^2)\frac{\log(f_i f_j)}{f_i f_j - \dfrac{\mu^2}{f_i f_j} + \mu^2 - 1}.$$

Let $a_x := \eta\|H(x)\|$. Since $|\eta\lambda_i| \le a_x$, the previous Lemma 1 gives $|K_{ij}-1| \le \delta_\mu(a_x)$. We may therefore write

$$K = \mathbf{1}\mathbf{1}^\top + E, \qquad |E_{ij}| \le \delta_\mu(a_x).$$

Using the definition of S,

$$S = \frac{\eta}{1+\mu^2}\widehat{\Sigma} \odot K = \frac{\eta}{1+\mu^2}\left(\widehat{\Sigma} + \widehat{\Sigma} \odot E\right).$$

Consequently,

$$\mathcal{D}(x) = \frac{\eta}{1+\mu^2} U\left(\widehat{\Sigma} + \widehat{\Sigma} \odot E\right) U^\top.$$

By Lemma 14 in HA-SME,

$$\|\widehat{\Sigma} \odot E\| \le \sqrt{d}\,\|\widehat{\Sigma}\| \max_{i,j} |E_{ij}|.$$

Since U is orthogonal, $\|\widehat{\Sigma}\| = \lambda_{\max}(\Sigma(x))$, therefore

$$\|\widehat{\Sigma} \odot E\| \le \sqrt{d}\,\lambda_{\max}(\Sigma(x))\delta_\mu(a_x).$$

Similarly to the proof of Lemma 3 in HA-SME, to prove that $\widehat{\Sigma} + \widehat{\Sigma} \odot E$ is positive semidefinite, it is sufficient to show that

$$\lambda_{\min}(\Sigma(x)) = \lambda_{\min}(\widehat{\Sigma}) \ge \|\widehat{\Sigma} \odot E\|.$$

Therefore, we need:

$$\lambda_{\min}(\Sigma(x)) \ge \sqrt{d}\,\lambda_{\max}(\Sigma(x))\delta_\mu(a_x),$$

which is precisely the second condition. Hence $\widehat{\Sigma} + \widehat{\Sigma} \odot E \succeq 0$, and consequently

$$S \succeq 0 \qquad \text{and} \qquad \mathcal{D}(x) = USU^\top \succeq 0.$$

□

Now we combine Lemma 1, Lemma 2 to establish sufficient conditions under which the SDE has unique strong solution, following the proof of Theorem 2 from HA-SME.

Theorem. Fix $0 \le \mu < 1$, and let $H(x) := \nabla^2 f(x)$. Assume that

$$f \in C_b^3(\mathbb{R}^d), \qquad \Sigma \in C_b^1(\mathbb{R}^d; \mathbb{R}^{d\times d}),$$

and that $\Sigma(x)$ is symmetric and positive definite for every $x \in \mathbb{R}^d$. Assume also that the stepsize satisfies the conditions of Section II.3 with $n = 1$, so that the drift b and the covariance matrix $\mathcal{D}$ of the forward momentum process satisfy

$$b \in C_b^1(\mathbb{R}^d; \mathbb{R}^d), \qquad \mathcal{D} \in C_b^1(\mathbb{R}^d; \mathbb{R}^{d\times d}).$$

Suppose, in addition, that at least one of the following conditions holds:

1. $H(x)$ and $\Sigma(x)$ commute for every $x \in \mathbb{R}^d$, and

$$\eta \sup_{x\in\mathbb{R}^d} \|H(x)\| < R(\mu) := (1-\sqrt{\mu})^2;$$

2. for every $x \in \mathbb{R}^d$,

$$\eta\|H(x)\| < R(\mu) \text{ and } \lambda_{\min}(\Sigma(x)) > \sqrt{d}\,\lambda_{\max}(\Sigma(x))\delta_\mu(\eta\|H(x)\|)\,.$$

Then the momentum HA-SME

$$dX_t^+ = b(X_t^+)\,dt + D(X_t^+)\,dW_t^+$$

has a unique global strong solution for every initial condition $X_0^+ \in \mathbb{R}^d$.

Proof. We first show that $\mathcal{D}(x) \succ 0 \quad$ for every $x \in \mathbb{R}^d$. Suppose first that condition 1 holds. Since $H(x)$ and $\Sigma(x)$ are real symmetric and commute, they can be simultaneously diagonalized. By the argument in the proof of Lemma 2, in this common eigenbasis the eigenvalues of $\mathcal{D}(x)$ are

$$\eta\,\widetilde{\lambda}_i(x)\,q_\mu\big(\eta\lambda_i(x)\big), \qquad i = 1, \dots, d,$$

where $\widetilde{\lambda}_i(x)$ are the eigenvalues of $\Sigma(x)$, and

$$q_\mu(z) := \frac{\log\big(f(\mu,z)^2\big)}{f(\mu,z)^2 - \dfrac{\mu^2}{f(\mu,z)^2} + \mu^2 - 1} > 0 \qquad \text{for } |z| < R(\mu).$$

Since $\Sigma(x) \succ 0$, we have $\widetilde{\lambda}_i(x) > 0$, while the stepsize condition gives $|\eta\lambda_i(x)| < R(\mu)$. Therefore every eigenvalue of $\mathcal{D}(x)$ is strictly positive, and hence $\mathcal{D}(x) \succ 0 \quad$ for every $x \in \mathbb{R}^d$.

Suppose instead that condition 2 holds. By the argument in the proof of Lemma 2,

$$\lambda_{\min}\big(\mathcal{D}(x)\big) \geq \frac{\eta}{1+\mu^2}\left[\lambda_{\min}(\Sigma(x)) - \sqrt{d}\,\lambda_{\max}(\Sigma(x))\delta_\mu(\eta\|H(x)\|)\right].$$

The strict inequality in condition 2 therefore implies that $\lambda_{\min}\big(\mathcal{D}(x)\big) > 0$. Hence $\mathcal{D}(x) \succ 0$ for every $x \in \mathbb{R}^d$.

We next prove that $D(x) := \sqrt{\mathcal{D}(x)}$ is locally Lipschitz. Fix $R > 0$, and let $K_R := \{x \in \mathbb{R}^d : \|x\| \leq R\}$. Since $\mathcal{D}$ is continuous and $\mathcal{D}(x) \succ 0$ for every x, the map $x \longmapsto \lambda_{\min}(\mathcal{D}(x))$ is continuous and strictly positive (by Weyl's inequality $|\lambda_{\min}(\mathcal{D}(x)) - \lambda_{\min}(\mathcal{D}(y))| \leq \|\mathcal{D}(x) - \mathcal{D}(y)\|$). Compactness of K_R therefore implies that $c_R := \min_{x\in K_R} \lambda_{\min}(\mathcal{D}(x)) > 0$. It follows that

$$\lambda_{\min}(D(x)) = \sqrt{\lambda_{\min}(\mathcal{D}(x))} \geq \sqrt{c_R}, \qquad x \in K_R.$$

Differentiating $D(x)^2 = \mathcal{D}(x)$ with respect to x_k yields

$$D(x)\frac{\partial D(x)}{\partial x_k} + \frac{\partial D(x)}{\partial x_k}D(x) = \frac{\partial \mathcal{D}(x)}{\partial x_k}.$$

After vectorization, like in the proof of Theorem 2 in HA-SME, this becomes

$$\operatorname{vec}\left(\frac{\partial D(x)}{\partial x_k}\right) = \big(D(x)\otimes I + I \otimes D(x)\big)^{-1} \operatorname{vec}\left(\frac{\partial \mathcal{D}(x)}{\partial x_k}\right).$$

Like in the proof of Theorem 2 in HA-SME, the eigenvalues of

$$D(x)\otimes I + I \otimes D(x)$$

are

$$\{\lambda_i(D(x)) + \lambda_j(D(x)) : 1 \leq i, j \leq d\}.$$

Thus, for $x \in K_R$,

$$\lambda_{\min}\left(D(x)\otimes I + I \otimes D(x)\right) \geq 2\sqrt{c_R},$$

and consequently

$$\left\|\big(D(x)\otimes I + I \otimes D(x)\big)^{-1}\right\| \leq \frac{1}{2\sqrt{c_R}}.$$

Since $\mathcal{D} \in C_b^1$, its first derivatives are bounded. It follows that, for every $1 \leq k \leq d$, $\sup_{x\in K_R}\left\|\frac{\partial D(x)}{\partial x_k}\right\| < \infty$. Therefore, D is Lipschitz on K_R. Since $R > 0$ was arbitrary, D is locally Lipschitz on $\mathbb{R}^d$.

Finally, $b \in C_b^1$, and hence b is globally Lipschitz and bounded. The matrix $\mathcal{D}$ is also bounded, so D is bounded as well. Thus, for some $C > 0$,

$$\|b(x)\| + \|D(x)\| \le C \le C(1 + \|x\|), \qquad x \in \mathbb{R}^d.$$

The coefficients b and D are therefore locally Lipschitz and satisfy the linear-growth condition. The standard existence and pathwise uniqueness theorem for Itô SDEs now shows that

$$dX_t^+ = b(X_t^+)\,dt + D(X_t^+)\,dW_t^+$$

has a unique global strong solution. □

III. Weak approximation error guarantees

1. Stochastic extensions of existing ODEs for modelling GD with momentum - heuristic motivations

We first consider the following two ODEs for GD with momentum studied by Kovachki and Stuart (2021) and Ghosh et al (2023), which achieve $O(\eta)$ and $O(\eta^2)$ discretization error, respectively.

$$\dot{x} = -\frac{\nabla f(x)}{1-\mu} \text{ and } \dot{x} = -\frac{\nabla f(x)}{1-\mu} - \eta\frac{1+\mu}{(1-\mu)^3}\frac{\nabla f(x)\cdot\nabla^2 f(x)}{2}.$$

Inspired by the construction of SME-1 and SME-2 and following the idea from section 3.1 of Li et al 2018, we will add a diffusion term of the form $\sqrt{\frac{\eta\Sigma}{c(\mu)}}$, where we will find an expression for $c(\mu)$ by comparing the discretization of the SDE and the discrete update steps. This derivation is not rigorous, just a heuristic motivation.

Consider SGDM

$$x_{k+1} = x_k - \eta\big(\nabla f(x_k) + \xi_k\big) + \mu(x_k - x_{k-1}),$$

where

$$\mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0, \qquad \mathbb{E}[\xi_k\xi_k^\top \mid \mathcal{F}_k] = \Sigma(x_k).$$

Define

$$R_k^{\mathrm{SGDM}} = x_{k+1} + \mu x_{k-1} - (1+\mu)x_k = -\eta\nabla f(x_k) - \eta\xi_k,$$

so

$$\mathbb{E}[R_k^{\mathrm{SGDM}} \mid \mathcal{F}_k] = -\eta\nabla f(x_k), \qquad \mathrm{Cov}(R_k^{\mathrm{SGDM}} \mid \mathcal{F}_k) = \eta^2\Sigma(x_k). \tag{1}$$

Now consider the candidate SDE

$$dX_t = b(X_t)\,dt + \sqrt{\frac{\eta}{c(\mu)}}\,\sigma(X_t)\,dW_t, \qquad \sigma\sigma^\top = \Sigma.$$

Its Euler–Maruyama discretization is

$$\widehat{X}_{m+1} = \widehat{X}_m + \eta b(\widehat{X}_m) + \frac{\eta}{\sqrt{c(\mu)}}\sigma(\widehat{X}_m)Z_m,$$

where $Z_m \overset{\text{i.i.d.}}{\sim} N(0, I)$.

Using the steps $m = k-1, k$,

$$\begin{aligned} R_k^{\mathrm{EM}} &:= \widehat{X}_{k+1} + \mu\widehat{X}_{k-1} - (1+\mu)\widehat{X}_k \\ &= \eta\Big[b(\widehat{X}_k) - \mu b(\widehat{X}_{k-1})\Big] \\ &\quad + \frac{\eta}{\sqrt{c(\mu)}}\Big[\sigma(\widehat{X}_k)Z_k - \mu\sigma(\widehat{X}_{k-1})Z_{k-1}\Big]. \end{aligned}$$

Locally,

$$b(\widehat{X}_{k-1}) \approx b(\widehat{X}_k), \qquad \sigma(\widehat{X}_{k-1}) \approx \sigma(\widehat{X}_k),$$

and hence

$$R_k^{\mathrm{EM}} \approx \eta(1-\mu)b(\widehat{X}_k) + \frac{\eta}{\sqrt{c(\mu)}}\sigma(\widehat{X}_k)(Z_k - \mu Z_{k-1}). \tag{2}$$

Matching the means in (1)–(2) gives

$$b(x) = -\frac{\nabla f(x)}{1-\mu}.$$

Since Z_k and Z_{k-1} are independent,

$$\mathrm{Cov}(Z_k - \mu Z_{k-1}) = (1+\mu^2)I.$$

Matching covariances gives

$$\frac{\eta^2(1+\mu^2)}{c(\mu)}\Sigma(x) = \eta^2\Sigma(x),$$

and therefore

$$\boxed{c(\mu) = 1+\mu^2.}$$

Note that this agrees with the following fact we showed in II.2

$$g_\mu(x,y) = \frac{\log\big(f(\mu,x)f(\mu,y)\big)}{f(\mu,x)f(\mu,y) - \dfrac{\mu^2}{f(\mu,x)f(\mu,y)} + \mu^2 - 1},$$

has an analytic extension to a neighborhood of $(0,0)$, with

$$g_\mu(0,0) = \frac{1}{1+\mu^2},$$

which will be the coefficient of $\eta\Sigma$ in the series of this function in the diffusion of the extension of HA-SME for momentum.

Therefore, the stochastic extensions are as follows:

$$\boxed{dX_t = -\frac{\nabla f(X_t)}{1-\mu}\,dt + \sqrt{\frac{\eta\Sigma(X_t)}{1+\mu^2}}dW_t.}$$

and

$$\boxed{\begin{aligned} dX_t = &\left[-\frac{\nabla f(X_t)}{1-\mu} - \eta\frac{1+\mu}{2(1-\mu)^3}\nabla^2 f(X_t)\nabla f(X_t)\right]dt \\ &+ \sqrt{\frac{\eta\Sigma(X_t)}{1+\mu^2}}dW_t. \end{aligned}}$$

2. Error analysis

For the analysis here we assume $f(x) \in C_b^5(\mathbb{R})$ and $\Sigma(x) \in C_b^3(\mathbb{R}^{d\times d})$, **as well as that Σ is uniformly elliptic**: $\Sigma(x) \succeq cI, \quad \forall x \in \mathbb{R}^d$. Moreover, assume that $\sup_{x,\varepsilon}\|\nabla F(x;\varepsilon)\| < \infty$.

In this section we will use the following notation from Krylov and Priola.

Notation.

Fix $0 < \alpha < 1$. For a bounded function $h : \mathbb{R}^d \to \mathbb{R}$, set $\|h\|_0 = \sup_{x\in\mathbb{R}^d}|h(x)|, \quad [h]_\alpha = \sup_{\substack{x,y\in\mathbb{R}^d \\ x\neq y}} \frac{|h(x)-h(y)|}{|x-y|^\alpha}$.

For an integer $k \geq 0$, $C_b^k = C_b^k(\mathbb{R}^d)$ denotes the space of functions h whose spatial derivatives $D^\gamma h$, $|\gamma| \leq k$, exist, are continuous, and are bounded. We use the norm $\|h\|_k = \sum_{|\gamma|\leq k}\|D^\gamma h\|_0$.

The space $C_b^{k+\alpha} = C_b^{k+\alpha}(\mathbb{R}^d)$ consists of the functions $h \in C_b^k$ such that all derivatives of order k are bounded α-Hölder continuous. Its norm is $\|h\|_{k+\alpha} = \sum_{|\gamma|\le k} \|D^\gamma h\|_0 + \sum_{|\gamma|=k} [D^\gamma h]_\alpha < \infty$ For brevity, we sometimes omit the b subscript in $C_b^{k+\alpha}$. For vector- and matrix-valued functions, we apply this entrywise and then sum: let $\|a\|_{k+\alpha} = \sum_{i,j=1}^d \|a^{ij}\|_{k+\alpha}$, $\quad \|b\|_{k+\alpha} = \sum_{i=1}^d \|b^i\|_{k+\alpha}$.

For $T \in [-\infty, \infty)$, put $\mathbb{R}_T^{d+1} = \{(t,x) \in \mathbb{R} \times \mathbb{R}^d : t \ge T\}$. We denote by $\mathcal{C}^{k+\alpha}(T)$ the set of functions $u = u(t,x)$ defined on $\mathbb{R}_T^{d+1}$ such that:

1. u is continuous on $\mathbb{R}_T^{d+1}$;
2. for every finite $t \ge T$, $u(t,\cdot) \in C_b^{k+\alpha}(\mathbb{R}^d)$, and $\sup_{t \ge T} \|u(t,\cdot)\|_{k+\alpha} < \infty$;
3. there exists a measurable function $g = g(t,x)$ on $\mathbb{R}_T^{d+1}$ such that, for every $\zeta \in C_0^\infty(\mathbb{R}^{d+1})$, the functions $\zeta(t,\cdot)g(t,\cdot)$ belong to $C_b^{k-2+\alpha}(\mathbb{R}^d)$, uniformly in t, and, for every $x \in \mathbb{R}^d$ and every finite s,t satisfying $T \le s < t$, one has $u(t,x) - u(s,x) = \int_s^t g(r,x)\,dr$. (For such a function u, we write $u_t = g$ and call g the generalized derivative of u with respect to t).

Let $T < S < \infty$. We define $\mathcal{C}^{k+\alpha}(T,S) = \left\{u \in \mathcal{C}^{k+\alpha}(T) : u(t,x) = u(S,x) \text{ for all } t \ge S,\ x \in \mathbb{R}^d\right\}$.

We claim that the stochastic extensions of the ODEs established by Kovachki and Stuart (2021) and Ghosh et al (2023), as well as our SDE from section II.2 that extends HA-SME, all achieve order 1 weak approximation. This will follow from the following

Main result. Let $\varphi \in C_b^5(\mathbb{R}^d)$ be a test function. Assume the SDE for $X(t)$, $t \in [0;T]$ with drift $b_\eta(x)$ and diffusion $\sqrt{\mathcal{D}_\eta(x)}$ satisfies the following conditions for some $\eta_0 > 0$:

- $\mathcal{D}_\eta \succeq k_\eta I, \quad k_\eta > 0$ for all $\eta_0 > \eta > 0$
- $b_\eta(x) = -\frac{\nabla f(x)}{1-\mu} + \eta F_\eta(\nabla^2 f(x))\nabla f(x)$ and $\mathcal{D}_\eta(x) = \eta G_\eta(\nabla^2 f(x), \Sigma(x))$, where F_η, G_η are (possibly truncated) series with $\|F_\eta(t)\|_\infty, \|G_\eta(t)\|_\infty \le O(1)$ for $0 < \eta < \eta_0$
- $\max_{0\le i\le d} \|[b_\eta(x)]_i\|_{C^3} \le O(1)$ for $0 < \eta < \eta_0$
- $\max_{0\le i,j\le d} \left\| \left[\mathcal{D}_\eta(x)\right]_{i,j} \right\|_{C^3} \le O(1)$ for $0 < \eta < \eta_0$

Let $0 \le N < \frac{T}{\eta}$. Then if $u^n(x) := \mathbb{E}(\varphi(x_n)|x_0 = x)$ and $u_\eta(t,x) := \mathbb{E}(\varphi(X(t))|X(0) = x)$, for all $x \in \mathbb{R}^d$ we have $\forall \eta \in (0;\eta_0)$:

$$\|u_\eta^N(x) - u_\eta(N\eta, x)\| = O(\eta).$$

Proof. For $0 \le n \le N$ define

$$\tau_n = (N-n)\eta,\ A_n(x) = \mathbb{E}\left[u(\tau_n, x_n) \,\middle|\, x_0 = x\right] \text{ and } D_n(x) = A_n(x) - A_{n-1}(x)$$

Observe that $A_0(x) = u(N\eta, x)$ and $A_N(x) = u^N(x)$. Moreover, if

$$R_n(x) := A_{n+1}(x) + \mu A_{n-1}(x) - (\mu+1)A_n(x) \iff D_{n+1} = \mu D_n + R_n, \text{ so}$$

$D_N + \ldots + D_1 = D_1 + (\mu D_1 + R_1) + (\mu^2 D_1 + \mu R_1 + R_2) + \ldots + (\mu^{N-1} D_1 + \mu^{N-2} R_1 + \mu^{N-3} R_2 + \ldots + R_{N-1})$.
Therefore, we have

$$u^N(x) - u(N\eta, x) = A_N - A_0 = \frac{1-\mu^N}{1-\mu} D_1 + \frac{1}{1-\mu} \sum_{j=1}^{N-1} \left(1 - \mu^{N-j}\right) R_j.$$

We will show that $D_1 = O(\eta)$ and $\forall R_n = O(\eta^2)$. This indeed would complete the proof, because

$$A_N - A_0 \le \frac{1}{1-\mu}(O(\eta) + (N-1)O(\eta^2)) = O(\eta),$$

since $N - 1 < \frac{T}{\eta}$.

We will use the following forms of Taylor expansions:

$$u(t+h_0, x+h_x) = u(t,x) + h_0 \partial_t u(t,x) + \nabla u(t,x) \cdot h_x + R_2(t,x;h).$$

where $h = (h_0, h_x)$,

$$R_2(t,x;h) = \int_0^1 (1-r) h^\mathsf{T} D^2_{t,x} u(t+rh_0, x+rh_x) h \, dr,$$

and

$$D_{t,x}u(t,x) = \begin{pmatrix} \partial_t u(t,x) \\ \nabla u(t,x) \end{pmatrix}, \; D^2_{t,x}u(t,x) = \begin{pmatrix} \partial_{tt} u(t,x) & (\partial_t \nabla u(t,x))^\mathsf{T} \\ \partial_t \nabla u(t,x) & \nabla^2 u(t,x) \end{pmatrix}.$$

We will require this remainder R_2 to be $O(\eta^2)$, for which we need $h = O(\eta)$ and all joint second derivatives of $u(t,x)$ are uniformly bounded on $t \in [0;T]$. For the former, it's sufficient to show $d_{k+1} := x_{k+1} - x_k = O(\eta)$. For D_1 to be $O(\eta)$, we require the first derivatives of $u(t,x)$ to be bounded (so we require the 6 entries of the above two matrices to be bounded). Assume for now all these hold and we will show them later.

Therefore, we have

$$u(\tau_n - \eta, x_n + d_{n+1}) = u(\tau_n, x_n) - \eta \partial_t u(\tau_n, x_n) + \nabla u(\tau_n, x_n) \cdot d_{n+1} + O(\eta^2),$$
$$u(\tau_n + \eta, x_n - d_n) = u(\tau_n, x_n) + \eta \partial_t u(\tau_n, x_n) - \nabla u(\tau_n, x_n) \cdot d_n + O(\eta^2).$$

Combining these two and taking expectation, we get

$$\begin{aligned} A_{n+1}(x) + \mu A_{n-1}(x) - (1+\mu) A_n(x) = \mathbb{E}\left[\nabla u(\tau_n, x_n) \cdot (d_{n+1} - \mu d_n) \,\middle|\, x_0 = x \right] \\ - \eta(1-\mu) \mathbb{E}\left[\partial_t u(\tau_n, x_n) \,\middle|\, x_0 = x \right] + O(\eta^2) \end{aligned}$$

$$R_n(x) = \; -\eta \, \mathbb{E}[\nabla f(x_n) \cdot \nabla u(\tau_n, x_n) \,|\, x_0 = x] - \eta(1-\mu) \, \mathbb{E}[\partial_t u(\tau_n, x_n) \,|\, x_0 = x] + O(\eta^2).$$

Since $\nabla u, \nabla^2 u$ and F_η, G_η are bounded, by the backward Kolmogorov equation we have:

$$\begin{aligned} \partial_t u(t,y) = Lu(t,y) = -\frac{\nabla f(y)}{1-\mu} \cdot \nabla u(t,y) + \eta F_\eta(\nabla^2 f(y)) \nabla f(y) \cdot \nabla u(t,y) \\ + \frac{1}{2} \eta G_\eta(\nabla^2 f(y), \Sigma(y)) : \nabla^2 u(t,y) = -\frac{\nabla f(y)}{1-\mu} \cdot \nabla u(t,y) + O(\eta) \end{aligned}$$

Setting $(t,y) = (\tau_n, x_n)$ above yields:

$$\begin{aligned} R_n(x) = \; -\eta \, \mathbb{E}[\nabla f(x_n) \cdot \nabla u(\tau_n, x_n) \,|\, x_0 = x] \\ - \eta \, \mathbb{E}[-\nabla f(x_n) \cdot \nabla u(\tau_n, x_n) + O(\eta) \,|\, x_0 = x] + O(\eta^2) = O(\eta^2). \end{aligned}$$

For D_1, similarly we have that

$$u(\tau_0 - \eta, x_0 - \eta \nabla F(x_0; \varepsilon)) = u(\tau_0, x_0) - \eta \partial_t u(\tau_0, x_0) - \eta \nabla u(\tau_0, x_0) \cdot \nabla F(x_0; \varepsilon) + O(\eta^2).$$

We apply again the backward Kolmogorov equation, but now the $O(\eta)$ terms don't cancel because after taking an expectation, their sum is $\eta(\frac{1}{1-\mu} \nabla f(x) - \nabla f(x)) \cdot \nabla u(\tau_0, x) + O(\eta^2)$. Thus, since $\nabla u, \nabla f$ are bounded, $D_1 = O(\eta)$.

Now we return to prove the required bounds:

Lemma 1. $d_n = O(\eta)$

Proof. By induction we have

$$d_{n+1} = \mu d_n - \eta \nabla F(x_n; \varepsilon_n) = \mu^2 d_{n-1} - \eta(\mu \nabla F(x_{n-1}; \varepsilon_{n-1}) + \nabla F(x_n; \varepsilon_n)) = \ldots = -\eta \sum_{k=0}^{n} \mu^k \nabla F(x_{n-k}; \varepsilon_{n-k}),$$

since $d_0 = 0$. Since all gradients of F are uniformly bounded, we have $\|d_{n+1}\| \le \frac{C\eta}{1-\mu}$. $\square$

Lemma 2. The first and second derivatives of $u(t,x)$ (i.e. the entries of $D_{t,x}u(t,x), D^2_{t,x}u(t,x)$) are uniformly bounded for $t \in [0;T]$ and $\eta \in (0;\eta_0)$ by a constant independent of η.

Proof. Let L be the generator of the SDE. From the proof of Lemma 17 from HA-SME, we have

$$\|\frac{\partial^n u(t,x)}{(\partial t)^n}\|_\infty \leq \|L^n \varphi\|_\infty.$$

Since the RHS contains derivatives of $b, \mathcal{D}$ up to order $2(n-1)$ and derivatives of φ up to order $2n$, for $n = 1, 2$ all of these are bounded by our assumptions, so the statement is true for $\partial_t u(t,x)$, $\partial_{tt} u(t,x)$.

Assuming we have shown a bound of the form

$$\sup_{0\leq t\leq T} \|\nabla u(t,x)\|_\infty \leq C_T \|\nabla \varphi\|_\infty,$$

since $\nabla \partial_t u(t,x) = \nabla L u(t,x) = \nabla e^{tL}(L\varphi(x)) = \nabla u(t, L\varphi(x))$, we get:

$$\sup_{0\leq t\leq T} \|\nabla \partial_t u(t,x)\|_\infty \leq C_T \|\nabla L\varphi\|_\infty \leq C\|\varphi\|_{C_b^3} = O(1).$$

Note that we don't require $\partial_t \nabla u(t,x) = \nabla \partial_t u(t,x)$, because boundedness of just $\nabla \partial_t u(t,x), \partial_{tt} u(t,x), \nabla^2 u(t,x)$ is sufficient. Indeed, note that we can bound the remainder of the order 1 Taylor series as follows:

$$\begin{aligned} R = &\ u(t+h_0, x+h_x) - u(t,x) - h_0 u_t(t,x) - \nabla u(t,x)\cdot h_x \\ = &\ \big[u(t+h_0, x+h_x) - u(t, x+h_x) - h_0 u_t(t, x+h_x)\big] \\ &+ [h_0\big(u_t(t,x+h_x) - u_t(t,x)\big)] \\ &+ \big[u(t,x+h_x) - u(t,x) - \nabla u(t,x)\cdot h_x\big]. \end{aligned}$$

Now we bounding the three terms as follows (via Taylor series for just one of the arguments and treating the other one as fixed) finishes the argument:

$$|u(t+h_0, x+h_x) - u(t, x+h_x) - h_0 u_t(t, x+h_x)| \leq \frac{1}{2}|h_0|^2 \|u_{tt}\|_\infty.$$

$$|h_0|\,|u_t(t,x+h_x) - u_t(t,x)| \leq |h_0|\,|h_x|\,\|\nabla u_t\|_\infty$$

$$|u(t,x+h_x) - u(t,x) - \nabla u(t,x)\cdot h_x| \leq \frac{1}{2}|h_x|^2 \|\nabla^2 u\|_\infty.$$

So, we are left to bound $\nabla u(t,x), \nabla^2 u(t,x)$ in terms of $\nabla\varphi, \nabla^2\varphi$. We will follow the ideas from Krylov and Priola's paper and will structure this into a few separate parts.

Part 1.

We will first establish the following generalization of Theorem 2.8 where α is shifted to $\alpha+2$. Its purpose is to establish that $u_\eta(t,x) \in C_b^{4+\alpha}(0,T)$ for a *fixed* η (this would be sufficient regularity to differentiate Kolmogorov's equation twice).

Theorem. Let $0 < \alpha < 1$, let $T \in [-\infty, \infty)$, and take and fix a finite $S > T$. Consider the operator

$$Lu = a^{ij}(t,x) D_{ij} u + b^i(t,x) D_i u - c(t,x) u.$$

Assume that the matrix $(a^{ij}(t,x))$ is symmetric and that the coefficients a^{ij}, b^i, and c are measurable in t. Suppose that there exist constants $\delta, K \in (0,\infty)$ such that $KI \geq (a^{ij}(t,x)) \geq \delta I, \quad c(t,x) \geq \delta$, for all $(t,x) \in [T,\infty)\times\mathbb{R}^d$, and $\sup_{t\geq T}(\|a(t,\cdot)\|_{2+\alpha} + \|b(t,\cdot)\|_{2+\alpha} + \|c(t,\cdot)\|_{2+\alpha}) \leq K$. Assume also that f is measurable in t, that $f(t,\cdot) \in C_b^{2+\alpha}(\mathbb{R}^d)$ for $t \in (T,S)$, and that $F_{2+\alpha} := \operatorname{ess\,sup}_{T<t<S} \|f(t,\cdot)\|_{2+\alpha} < \infty$.

Let $g \in C_b^{4+\alpha}(\mathbb{R}^d)$. Then there exists a unique $u \in \mathcal{C}^{4+\alpha}(T,S)$ satisfying $u_t + Lu = f \quad$ in $(T,S)\times\mathbb{R}^d$ and such that $u(S,x) = g(x), \quad x \in \mathbb{R}^d$. Moreover, for any finite $t \in [T,S]$, we have

$$\|u(t,\cdot)\|_{4+\alpha} \leq N\left(F_{2+\alpha} + \|g\|_{4+\alpha}\right),$$

where $N = N(\delta, K, \alpha, d)$.

Proof. We will show that the equivalents of Theorems 2.4 and 2.5 hold for our setup - then the same proof of Theorem 2.8. works here.

We first establish a result similar to Theorem 2.4. We claim that if $u \in \mathcal{C}^{4+\alpha}(T)$ satisfies $u_t + Lu = f$ for $t \geq T$, then $\|u(t,\cdot)\|_{4+\alpha} \leq NF_{2+\alpha}, \forall t \geq T$. We will show this by applying Theorem 2.4 consecutively three times for $u, v_k = D_k u, H_{kl} = D_{kl} u$. In particular, we will use the following version that doesn't involve F_0 :

$$w_t + Lw = h, \quad w \in \mathcal{C}^{2+\alpha}(T), \quad M_\alpha := \operatorname*{ess\,sup}_{t \geq T} \|h(t,\cdot)\|_\alpha < \infty \implies \sup_{t \geq T} \|w(t,\cdot)\|_{2+\alpha} \leq NM_\alpha.$$

(this version holds because we can take $F_0 = \delta^{-1} M_\alpha$ since $|h(t,x)| \leq M_\alpha \leq \delta^{-1} M_\alpha c(t,x)$ as $c \geq \delta$).

Clearly $\sup_{t \geq T} \|u(t,\cdot)\|_{2+\alpha} \leq N \operatorname{ess\,sup}_{t \geq T} \|f(t,\cdot)\|_\alpha \leq NF_{2+\alpha}$. Differentiating gives

$$(v_k)_t + Lv_k = D_k f - (D_k a^{ij}) D_{ij} u - (D_k b^i) D_i u + (D_k c) u := f_k.$$

Using Hölder's inequality $\|pq\|_\alpha \leq C \|p\|_\alpha \|q\|_\alpha$ and the uniform $C_b^{2+\alpha}$-bounds on the coefficients imply

$$\|f_k(t,\cdot)\|_\alpha \leq N \left(\|f(t,\cdot)\|_{1+\alpha} + \|u(t,\cdot)\|_{2+\alpha} \right) \implies \sup_{t \geq T} \|u(t,\cdot)\|_{3+\alpha} \leq NF_{2+\alpha},$$

where we used the previous bound and Theorem 2.4. for v_k.

Finally, differentiating twice gives $(H_{k\ell})_t + LH_{k\ell} = f_{k\ell}$, where

$$\begin{aligned} f_{k\ell} = {} & D_{k\ell} f - (D_{k\ell} a^{ij}) D_{ij} u - (D_k a^{ij}) D_{\ell ij} u - (D_\ell a^{ij}) D_{kij} u \\ & - (D_{k\ell} b^i) D_i u - (D_k b^i) D_{\ell i} u - (D_\ell b^i) D_{ki} u \\ & + (D_{k\ell} c) u + (D_k c) D_\ell u + (D_\ell c) D_k u. \end{aligned}$$

Similarly,

$$\|f_{k\ell}(t,\cdot)\|_\alpha \leq N \left(\|f(t,\cdot)\|_{2+\alpha} + \|u(t,\cdot)\|_{3+\alpha} \right) \implies \sup_{t \geq T} \|u(t,\cdot)\|_{4+\alpha} \leq NF_{2+\alpha}.$$

The version of Theorem 2.5 for $\alpha+4$ instead of $\alpha+2$ is that there exists unique $u \in C^{\alpha+4}(T)$ that satisfies $u_t + Lu = f$. The proof extends to this setup by similar arguments we used for Theorem 2.4 - usually we differentiate twice more and apply the corresponding results to the resulting higher derivatives, keeping the core arguments unchanged. $\square$

We now show how this result is applied in our situation. Using Remark 2.9, fix η and let

$$T := 0, \quad S := T, \quad g(x) = \varphi(x), \quad f(s,x) = 0, \quad a = \frac{1}{2} D_\eta, \quad b = b_\eta, \quad c(s,x) = 1.$$

Here T in the first equality denotes the one from the above theorem, the other one is from the main result. Since $\mathcal{D}_\eta$ is uniformly elliptic for fixed η, $b_\eta, D_\eta \in C_b^3 \subset C_b^{2+\alpha}$ and $\varphi \in C_b^5 \subset C_b^{4+\alpha}$, the assumptions are satisfied.

We claim that the unique $u(t,x)$ is in fact $\hat{v}_\eta(t,x) = e^{s-T} u_\eta(T-s,x)$ for $0 \leq s \leq T$ - this will allow us to establish a regularity condition on u_η. Indeed, $\hat{v}_\eta \in C^{4+\alpha}(0,T)$ satisfies

$$\partial_s \widehat{v}_\eta + L_\eta \widehat{v}_\eta - \widehat{v}_\eta = 0, \quad \widehat{v}_\eta(T,\cdot) = \varphi,$$

so if $v_\eta(t,x) = e^t \widehat{v}_\eta(T-t,x)$, we have $\partial_t v_\eta = L_\eta v_\eta,\ v_\eta(0,\cdot) = \varphi$. Using this along with Ito's lemma yields:

$$d\, v_\eta(t-r, X_r) = \nabla v_\eta(t-r, X_r)^\mathsf{T} D_\eta(X_r)\, dW_r \implies \varphi(X_t) = v_\eta(t,x) + \int_0^t \nabla v_\eta(t-r, X_r)^\mathsf{T} D_\eta(X_r)\, dW_r.$$

Taking expectation gives $u_\eta(t,x) = \mathbb{E}_x[\varphi(X_t)] = v_\eta(t,x)$. Indeed, since $\hat{v}_\eta \in C^{4+\alpha}(0,T)$ means $v_\eta \in C^{4+\alpha}(0,T)$, we have

$$\begin{aligned} \mathbb{E}_x \int_0^t \left| D_\eta(X_r)^\mathsf{T} \nabla v_\eta(t-r, X_r) \right|^2 dr = \mathbb{E}_x \int_0^t \nabla v_\eta(t-r, X_r)^\mathsf{T} \mathcal{D}_\eta(X_r) \nabla v_\eta(t-r, X_r)\, dr \\ \leq t \|\mathcal{D}_\eta\|_\infty \sup_{0 \leq s \leq t} \|\nabla v_\eta(s,\cdot)\|_\infty^2 < \infty, \end{aligned}$$

so if $G_r = D_\eta(X_r)^\mathsf{T}\nabla v_\eta(t-r, X_r)$, then it's well-known *(the Itô integral of a predictable square-integrable process is a continuous square-integrable martingale)* that $\int_0^t G_s\,dW_s, \quad 0 \le t \le T$ is a continuous square-integrable $(\mathcal{F}_t)$-martingale, which means that for $0 \le q \le s \le T$,

$$\mathbb{E}\left[\int_0^s G_r\,dW_r \middle| \mathcal{F}_q\right] = \int_0^q G_r\,dW_r \implies \mathbb{E}\left[\int_0^s G_r\,dW_r \middle| \mathcal{F}_0\right] = 0,$$

so we have

$$\mathbb{E}\left[\int_0^s G_r\,dW_r\right] = \mathbb{E}\left[\mathbb{E}\left[\int_0^s G_r\,dW_r \middle| \mathcal{F}_0\right]\right] = 0.$$

Therefore,

$$u_\eta(t,\cdot) \in C_b^{4+\alpha}, \quad \partial_t u_\eta(t,\cdot) \in C_b^{2+\alpha}, \quad \forall t \in [0;T].$$

Thus, we also have

$$V_\eta(t,x) = |\nabla u_\eta(t,x)|^2 \in C_b^{2+\alpha}(0,T), \quad W_\eta(t,x) = |\nabla^2 u_\eta(t,x)|_F^2 \in C_b^{2+\alpha}(0,T), \quad (V_\eta)_t \in C_b^{1+\alpha} \subset C_b^\alpha, \quad (W_\eta)_t \in C_b^\alpha.$$

Part 2.

Now we will derive the following corollary of theorem 4.1.

Corollary of Theorem 4.1. Let $T > 0$ and, in $[0,T] \times \mathbb{R}^d$, consider the operator

$$\mathscr{L}_t F(t,x) = a^{ij}(t,x) D_{ij} F(t,x) + b^i(t,x) D_i F(t,x).$$

Assume that:

(i) The functions a^{ij} and b^i are measurable, the matrix $(a^{ij}(t,x))$ is symmetric and nonnegative definite, and

$$|a(t,x)| \le K(t)(1+|x|^2), \qquad |b(t,x)| \le K(t)(1+|x|)$$

in $[0,T] \times \mathbb{R}^d$, where $K(t)$ is a locally bounded function on $[0,T]$.

(ii) The function $g : [0,T] \to [0,\infty)$ is measurable and locally integrable.

(iii) We are given a function $F \in \mathcal{C}^{2+\alpha}(0,T)$.

(iv) There is a constant $C \in \mathbb{R}$ such that $(\partial_t - \mathscr{L})F(t,x) \le CF(t,x) + g(t), \ \forall (t,x) \in [0,T] \times \mathbb{R}^d$

Then, in $[0,T] \times \mathbb{R}^d$,

$$\boxed{F(t,x) \le e^{Ct}\|F(0,\cdot)\|_0 + \int_0^t e^{C(t-r)} g(r)\,dr}.$$

Proof. Let $B(t) = e^{Ct}\|F(0,\cdot)\|_0 + \int_0^t e^{C(t-r)} g(r)\,dr$, which satisfies $B'(t) = CB(t) + g(t)$, and let $R(t,x) = F(t,x) - B(t)$. Then condition (iv) rewrites as $(\partial_t - \mathscr{L}_t)R \le CR, \quad R(0,x) \le 0$.

Now we will specify the choices we use to apply Theorem 4.1. We work on $[0,\infty) \times \mathbb{R}^d$ (so $T := 0$ in the notation of Theorem 4.1) and let

$$u(s,x) = \Psi(s,x) := e^{(C+1)(s-T)} R(T-s,x).$$

for $s \in [0;T]$ and let $\Psi(s,x) = \Psi(T,x) = R(0,x)$ for $s \ge T$. For $0 \le s < T$, choose

$$\widetilde{a}^{ij}(s,x) = a^{ij}(T-s,x), \qquad \widetilde{b}^i(s,x) = b^i(T-s,x), \qquad \widetilde{c}(s,x) = 1.$$

and for $s \ge T$ let those be $0, 0, 1$, respectively. Let $\delta = 1, F_0 = 0$ and

$$\widetilde{f}(s,x) = \partial_s \Psi + \widetilde{a}^{ij} D_{ij}\Psi + \widetilde{b}^i D_i \Psi - \Psi \ge 0 = -F_0 \widetilde{c}(s,x).$$

To check the last inequality, note that for $s > T$, $\partial_s \Psi = \widetilde{a}^{ij} D_{ij}\Psi = \widetilde{b}^i D_i \Psi = 0$ and $-\Psi(T,x) = -R(0,x) \ge 0$, while for $s \le T$ we have $\partial_s \Psi(s,x) = e^{(C+1)(s-T)}\left[(C+1)R - \partial_t R\right](T-s,x)$, so:

$$\widetilde{f}(s,x) = e^{(C+1)(s-T)}\left[(C+1)R - \partial_t R + \mathscr{L}_t R - R\right]_{t=T-s} = e^{(C+1)(s-T)}\left[CR - (\partial_t - \mathscr{L}_t)R\right](T-s,x) \ge 0$$

Clearly $\widetilde{a}^{ij}, \widetilde{b}^i, \widetilde{c}, \widetilde{f}$ satisfy the conditions in Theorem 4.1 because time-reversal preserves (i), and (ii) implies $\widetilde{f}$ is measurable and locally integrable in t and we showed above that the choices of $\delta = 1, F_0 = 0$ work. The third condition on $u(s,x)$ holds because (iii) implies that $u \in C^{2+\alpha}(0)$. Finally, (iv) holds because $\Psi_s + \widetilde{L}_s\Psi = \widetilde{f}, \quad \widetilde{L}_s = \widetilde{a}^{ij}(s,x)D_{ij} + \widetilde{b}^i(s,x)D_i - 1$.

Therefore, we get

$$\Psi(s,x) \le 0 \quad \text{for all } s \ge 0,\ x \in \mathbb{R}^d.$$

For any $t \in [0;T]$, setting $s = T - t$ gives $F(t,x) \le B(t)$, as desired. $\square$

Part 3. Now we will establish bounds in the form of (iv) for $F = V_\eta, W_\eta$ in the above corollary, by differentiating Kolmogorov's backward equation. *All differentiations are well-defined because of the regularity conditions established in Part 1.*

We will first establish the following estimate:

Claim. For every $x \in \mathbb{R}^d$, every $r \in \{1,\dots,d\}$, and every symmetric matrix S, there exists constant C_d such that

$$|(\partial_r \mathcal{D}_\eta(x)) : S|^2 \le C_d \|\mathcal{D}_\eta\|_{C_b^2} \operatorname{tr}(S\mathcal{D}_\eta(x)S)\,.$$

Remark. Everywhere in this section, C_d is some constant depending on d, not necessarily the same everywhere (to avoid using different notations for different constants, as this doesn't matter)

Proof. Since S is symmetric, there are an orthonormal basis $q_1,\dots,q_d$ and real numbers $\lambda_1,\dots,\lambda_d$ such that $S = \sum_{i=1}^d \lambda_i q_i q_i^\mathsf{T}$. Let $A := \partial_r \mathcal{D}_\eta(x), \quad \mathcal{D} := \mathcal{D}_\eta(x)$; we have $A : S = \operatorname{tr}(AS) = \sum_{i=1}^d \lambda_i \operatorname{tr}(Aq_iq_i^\mathsf{T}) = \sum_{i=1}^d \lambda_i q_i^\mathsf{T} A q_i$, We claim that $|q^\top A q|^2 \le C_d\|\mathcal{D}_\eta\|_{C_b^2} q^\top D q$ for any unit $q \in \mathbb{R}^d$. Assuming this, then by Cauchy Schwarz's inequality we have

$$\begin{aligned}|A:S| &\le \sqrt{C_d\|\mathcal{D}_\eta\|_{C_b^2}} \sum_{i=1}^d |\lambda_i| \sqrt{q_i^\top \mathcal{D} q_i} \le \sqrt{C_d\|\mathcal{D}_\eta\|_{C_b^2}} \left(\sum_{i=1}^d 1\right)^{1/2} \left(\sum_{i=1}^d \lambda_i^2 q_i^\top \mathcal{D} q_i\right)^{1/2} \\ &= \sqrt{C_d\|\mathcal{D}_\eta\|_{C_b^2}} \left(\sum_{i=1}^d \lambda_i^2 q_i^\top \mathcal{D} q_i\right)^{1/2} = \sqrt{C_d\|\mathcal{D}_\eta\|_{C_b^2} \operatorname{tr}(S\mathcal{D}S)}.\end{aligned}$$

as desired.

Let $h(s) := q^\top \mathcal{D}_\eta(x+se_r)q \ge 0$ and note that $h'(s) = q^\top(\partial_r \mathcal{D}_\eta)(x+se_r)q, \quad h''(s) = q^\top(\partial_{rr}\mathcal{D}_\eta)(x+se_r)q$ with $|h''(s)| \le C_d\|\mathcal{D}_\eta\|_{C_b^2}$. and $h''(s)$ is continuous since $\mathcal{D}_\eta \in C_b^3$. The Glaeser inequality $|f'(x)|^2 \le 2\|f''\|_\infty f(x)$ therefore gives

$$\left|q^\top(\partial_r \mathcal{D}_\eta(x))q\right|^2 = |h'(0)|^2 \le C_d\|\mathcal{D}_\eta\|_{C_b^2} h(0) = C_d\|\mathcal{D}_\eta\|_{C_b^2} q^\top \mathcal{D}_\eta(x) q. \quad \square$$

Part 3.1. $F = V_\eta$

Now let $v_i := \partial_i u_\eta, \quad H := \nabla^2 u_\eta$. Differentiating Kolmogorov's equation $\partial_t u_\eta = L_\eta u_\eta$ with respect to x_i gives

$$\partial_t v_i = b_\eta^k \partial_k v_i + \frac{1}{2}\mathcal{D}_\eta^{k\ell}\partial_{k\ell} v_i + (\partial_i b_\eta^k)v_k + \frac{1}{2}(\partial_i \mathcal{D}_\eta^{k\ell})H_{k\ell} \iff (\partial_t - L_\eta)v_i = (\partial_i b_\eta^k)v_k + \frac{1}{2}(\partial_i \mathcal{D}_\eta) : H.$$

$$\implies (\partial_t - L_\eta)V_\eta = 2v_i(\partial_i b_\eta^k)v_k + v_i(\partial_i \mathcal{D}_\eta) : H - \sum_{i=1}^d (\nabla v_i)^\top \mathcal{D}_\eta \nabla v_i.$$

Note that the last term is $\operatorname{tr}\big(H\mathcal{D}_\eta H\big) \ge 0$ (because ∇v_i is the i-th row of H and $\mathcal{D}_\eta$ is positive definite). Clearly we have $\left|2v_i(\partial_i b_\eta^k)v_k\right| \le C_d\|\nabla b_\eta\|_\infty |v|^2 = C_d\|\nabla b_\eta\|_\infty V_\eta$. Moreover, by Cauchy-Schwarz, the Claim and AM-GM inequality we get:

$$\left|\sum_{i=1}^d v_i(\partial_i \mathcal{D}_\eta) : H\right| \le |v| \left(\sum_{i=1}^d |(\partial_i \mathcal{D}_\eta) : H|^2\right)^{1/2} \le C_d\|\mathcal{D}_\eta\|_{C_b^2}^{1/2}|v|\sqrt{\operatorname{tr}\big(H\mathcal{D}_\eta H\big)} \le \frac{1}{2}\operatorname{tr}\big(H\mathcal{D}_\eta H\big) + C_d\|\mathcal{D}_\eta\|_{C_b^2}|v|^2.$$

Combining all these gives

$$(\partial_t - L_\eta)V_\eta \le C_d\left(\|\nabla b_\eta\|_\infty + \|\mathcal{D}_\eta\|_{C_b^2}\right)V_\eta - \frac{1}{2}\operatorname{tr}\big(H\mathcal{D}_\eta H\big) \le C_{1,\eta}V_\eta, \quad C_{1,\eta} := C_d\left(\|\nabla b_\eta\|_\infty + \|\mathcal{D}_\eta\|_{C_b^2}\right)$$

By the corollary of Theorem 4.1 for $F = V_\eta$ (which satisfies (iii) by Part 1 and (iv) by the above inequality) we get

$$V_\eta(t,x) \le e^{C_{1,\eta}t}\|V_\eta(0,\cdot)\|_\infty = e^{C_{1,\eta}t}\|\nabla\varphi\|_\infty \iff \|\nabla u_\eta(t,\cdot)\|_\infty \le e^{C_{1,\eta}t/2}\|\nabla\varphi\|_\infty.$$

But clearly $C_{1,\eta} \le C_d\left(\|b_\eta\|_{C_b^2} + \|\mathcal{D}_\eta\|_{C_b^2}\right) \le C_d \sup_{0<\eta\le\eta_0}\left(\|b_\eta\|_{C_b^2} + \|\mathcal{D}_\eta\|_{C_b^2}\right) < C_1$ for $\eta \in (0;\eta_0)$, where C_1 is now independent of η. because of our assumptions on $b_\eta, \mathcal{D}_\eta$. Therefore

$$\boxed{\sup_{0<\eta\le\eta_0}\sup_{0\le t\le T}\|\nabla u_\eta(t,\cdot)\|_\infty \le e^{C_1T/2}\|\nabla\varphi\|_\infty = O(1)}$$

Part 3.2. $F = W_\eta$

Let $H_{ij} = \partial_{ij}u_\eta$, $W_\eta = \sum H_{ij}^2$; differentiating $\partial_t u_\eta = b_\eta^k\partial_k u_\eta + \frac{1}{2}\mathcal{D}_\eta^{k\ell}\partial_{k\ell}u_\eta$ with respect to x_i, x_j gives:

$$(\partial_t - L_\eta)H_{ij} = (\partial_i b_\eta^k)H_{kj} + (\partial_j b_\eta^k)H_{ki} + (\partial_{ij}b_\eta^k)v_k + \frac{1}{2}(\partial_i\mathcal{D}_\eta) : T_j + \frac{1}{2}(\partial_j\mathcal{D}_\eta) : T_i + \frac{1}{2}(\partial_{ij}\mathcal{D}_\eta) : H := G_{ij}$$

where T_j is a symmetric matrix with $(T_j)_{k\ell} := \partial_{jk\ell}u_\eta$. Therefore

$$(\partial_t - L_\eta)W_\eta = 2\sum_{i,j} H_{ij}G_{ij} - \sum_{i,j}(\nabla H_{ij})^\top\mathcal{D}_\eta\nabla H_{ij} \iff$$

$$\begin{aligned}(\partial_t - L_\eta)W_\eta = {} & 2H_{ij}\left[(\partial_i b_\eta^k)H_{kj} + (\partial_j b_\eta^k)H_{ki}\right] + 2H_{ij}(\partial_{ij}b_\eta^k)v_k \\ & + H_{ij}(\partial_i\mathcal{D}_\eta) : T_j + H_{ij}(\partial_j\mathcal{D}_\eta) : T_i + H_{ij}(\partial_{ij}\mathcal{D}_\eta) : H - \sum_{i,j}(\nabla H_{ij})^\top\mathcal{D}_\eta\nabla H_{ij}.\end{aligned}$$

where the last term is $Q_\eta := \sum_{j=1}^d \operatorname{tr}(T_j\mathcal{D}_\eta T_j) \ge 0$. Note that

$$\left|2H_{ij}\left[(\partial_i b_\eta^k)H_{kj} + (\partial_j b_\eta^k)H_{ki}\right]\right| \le C_d\|\nabla b_\eta\|_\infty|H|_F^2;$$

$$\left|2H_{ij}(\partial_{ij}b_\eta^k)v_k\right| \le C_d\|\nabla^2 b_\eta\|_\infty|H|_F|v| \le C_d\|\nabla^2 b_\eta\|_\infty\big(|H|_F^2 + |v|^2\big).$$

By Cauchy-Schwarz we have:

$$\left|\sum_{i,j} H_{ij}(\partial_{ij}\mathcal{D}_\eta) : H\right| \le \sum_{i,j}|H_{ij}|\,|\partial_{ij}\mathcal{D}_\eta|_F\,|H|_F \le \left(\max_{i,j}|\partial_{ij}\mathcal{D}_\eta|_F\right)|H|_F\sum_{i,j}|H_{ij}|$$

$$\le d\left(\max_{i,j}|\partial_{ij}\mathcal{D}_\eta|_F\right)|H|_F^2 \le C_d\|\mathcal{D}_\eta\|_{C_b^2}|H|_F^2.$$

because

$$|(\partial_{ij}\mathcal{D}_\eta) : H| \le |\partial_{ij}\mathcal{D}_\eta|_F\,|H|_F, \quad \sum_{i,j}|H_{ij}| \le \left(\sum_{i,j}1\right)^{1/2}\left(\sum_{i,j}|H_{ij}|^2\right)^{1/2} = d\,|H|_F.$$

Finally, by using again Cauchy-Schwarz, the Claim and AM-GM, we bound the remaining two terms:

$$\left|\sum_{i,j} H_{ij}\Big[(\partial_i\mathcal{D}_\eta) : T_j + (\partial_j\mathcal{D}_\eta) : T_i\Big]\right| = 2\left|\sum_{i,j} H_{ij}(\partial_i\mathcal{D}_\eta) : T_j\right| \le 2|H|_F\left(\sum_{i,j}|(\partial_i\mathcal{D}_\eta) : T_j|^2\right)^{1/2}$$

$$\le C_d\|\mathcal{D}_\eta\|_{C_b^2}^{1/2}|H|_F\left(\sum_j \operatorname{tr}(T_j\mathcal{D}_\eta T_j)\right)^{1/2} = C_d\|\mathcal{D}_\eta\|_{C_b^2}^{1/2}|H|_F Q_\eta^{1/2} \le \frac{1}{2}Q_\eta + C_d\|\mathcal{D}_\eta\|_{C_b^2}|H|_F^2.$$

Combining all four inequalities yields:

$$(\partial_t - L_\eta)W_\eta \leq -\frac{1}{2}Q_\eta + C_d\Big(\|\mathcal{D}_\eta\|_{C_b^2} + \|\nabla b_\eta\|_\infty + \|\nabla^2 b_\eta\|_\infty + \|\nabla^2 \mathcal{D}_\eta\|_\infty\Big)|H|_F^2 + C_d\|\nabla^2 b_\eta\|_\infty |v|^2.$$

$$\implies (\partial_t - L_\eta)W_\eta \leq C_{2,\eta}W_\eta + C_{3,\eta}V_\eta \implies (\partial_t - L_\eta)W_\eta \leq C_{2,\eta}W_\eta + C_{3,\eta}\|V_\eta(t,\cdot)\|_\infty.$$

By the corollary of Theorem 4.1 for $F = W_\eta$ (as (iii) holds due to Part 1. and (iv) due to the inequality above) we get

$$\|\nabla^2 u_\eta(t,\cdot)\|_\infty^2 \leq e^{C_{2,\eta}t}\|\nabla^2\varphi\|_\infty^2 + C_{3,\eta}\int_0^t e^{C_{2,\eta}(t-r)}\|\nabla u_\eta(r,\cdot)\|_\infty^2\, dr.$$

Finally, using the previous bound $\|\nabla u_\eta(r,\cdot)\|_\infty^2 \leq e^{C_{1,\eta}r}\|\nabla\varphi\|_\infty^2$, we arrive at

$$\boxed{\|\nabla^2 u_\eta(t,\cdot)\|_\infty^2 \leq e^{C_{2,\eta}t}\|\nabla^2\varphi\|_\infty^2 + C_{3,\eta}\|\nabla\varphi\|_\infty^2\int_0^t e^{C_{2,\eta}(t-r)+C_{1,\eta}r}\, dr.}$$

As $C_{2,\eta}, C_{3,\eta}$ are uniformly bounded for $\eta \in (0;\eta_0), t \in [0;T]$, $\sup_{0<\eta\leq\eta_0}\sup_{0\leq t\leq T}\|\nabla^2 u_\eta(t,\cdot)\|_\infty \leq O(1)$. This finishes the proof of Lemma 2 and the main result. ■

Now we show how this result applies for our SDEs.

Part 1. - verification for the truncated SDEs

Let

$$b_\eta^{(1)}(x) = -\frac{1}{1-\mu}\nabla f(x), \quad b_\eta^{(2)}(x) = -\frac{1}{1-\mu}\nabla f(x) - \eta a_\mu \nabla^2 f(x)\nabla f(x), \quad \mathcal{D}_\eta(x) = \frac{\eta}{1+\mu^2}\Sigma(x).$$

The first requirement clearly holds with $k_\eta = \frac{c\eta}{\mu^2+1}$. The second condition also holds as $f \in C_b^5, \Sigma \in C_b^3$. The other two requirements also hold since:

$$\max_{1\leq i\leq d}\big\|[b_\eta^{(1)}]_i\big\|_{C_b^3} \leq \frac{C_d}{1-\mu}\|f\|_{C_b^4}, \quad \max_{1\leq i\leq d}\big\|[b_\eta^{(2)}]_i\big\|_{C_b^3} \leq C_{\mu,d}\left(\|f\|_{C_b^4} + \eta_0\|f\|_{C_b^5}^2\right),$$

$$\max_{1\leq i,j\leq d}\big\|[\mathcal{D}_\eta]_{ij}\big\|_{C_b^3} \leq \frac{\eta_0}{1+\mu^2}\max_{1\leq i,j\leq d}\big\|[\Sigma]_{ij}\big\|_{C_b^3}.$$

Part 2. - verification for momentum HA-SME

To verify the first requirement, we will take two cases, similarly to the well-posedness result in II.4.

If $H(x), \Sigma(x)$ commute, recall from the proof of the Theorem in II.4 that the eigenvalues of $\mathcal{D}_\eta$ are $\eta\,\widetilde{\lambda}_i(x)\,q_\mu(\eta\lambda_i(x))$, so $\mathcal{D}_\eta(x) \succeq \eta\, c\, q_{\mu,*} I \quad$ for all $x \in \mathbb{R}^d, \quad 0 < \eta \leq \eta_0$, because q_μ is continuous and positive on compact domain $[-a;a]$ with positive removable singularity at 1, meaning that $q_{\mu,*} := \inf_{|z|\leq a} q_\mu(z) > 0$.

For the other case, the eigenvalues are lower bounded by

$$\frac{\eta}{1+\mu^2}\left[\lambda_{\min}(\Sigma(x)) - \sqrt{d}\,\lambda_{\max}(\Sigma(x))\delta_\mu(\eta\|H(x)\|)\right] \geq \frac{\eta}{1+\mu^2}\left[c - \sqrt{d}\,c'\,\delta_\mu(\eta L)\right],$$

where $L = \sup\|H(x)\|$ and c' is an upper bound on the eigenvalues of $\Sigma(x) \in C_b^2$. Since $\delta_\mu(a) \longrightarrow 0 \quad$ as $a \downarrow 0$, the second factor is lower bounded by $\frac{c}{2}$ for $\eta \in (0;\eta_0)$

The third and fourth requirements directly hold due to the results from section II.3 since $f \in C_b^5$.

For the second requirement, let

$$c_\mu(z) := \frac{\log f(\mu,z)}{z}, \qquad c_\mu(0) = -\frac{1}{1-\mu}, \quad b_\eta(x) = c_\mu(\eta H(x))\,\nabla f(x).$$

Define

$$\psi_\mu(z) := \begin{cases} \dfrac{c_\mu(z) - c_\mu(0)}{z}, & z \neq 0, \\ c_\mu'(0), & z = 0. \end{cases}$$

We have that c_μ is analytic when $|z| < (1-\sqrt{\mu})^2$, so ψ_μ is also analytic near zero. Then

$$\begin{aligned} b_\eta(x) &= c_\mu(0)\nabla f(x) + \big(c_\mu(\eta H(x)) - c_\mu(0)I\big)\nabla f(x) \\ &= -\frac{\nabla f(x)}{1-\mu} + \eta\,\psi_\mu(\eta H(x))H(x)\nabla f(x). \end{aligned} \qquad \Longrightarrow\ F_\eta(H) = \psi_\mu(\eta H)H.$$

Then if $\lambda := \sup_x \|H(x)\| < \infty$ and $\eta\lambda \le a < (1-\sqrt{\mu})^2$, we have

$$\sup_x \|F_\eta(H(x))\| \le \lambda \sup_{|z|\le a} |\psi_\mu(z)| = O(1).$$

For the diffusion, we have

$$G_\eta(H,\Sigma) = \sum_{p,q\ge 0} a_{p,q}^{(\mu)} \eta^{p+q} H^p \Sigma H^q, \quad g_\mu(x,y) = \sum_{p,q\ge 0} a_{p,q}^{(\mu)} \eta^{p+q} x^p y^q, \quad a_{0,0}^{(\mu)} = 1/(1+\mu^2)$$

so if $S := \sup_x \|\Sigma(x)\| < \infty$ and $\eta\lambda \le a < R_\mu$ (where $R_\mu = (1-\sqrt{\mu})^2$ is its radius of convergence defined in the second Lemma in II.3), then similarly ($\eta\lambda$ lies in a fixed compact subset of the convergence domain)

$$\sup_x \|G_\eta(H(x),\Sigma(x))\| \le S \sum_{p,q\ge 0} |a_{p,q}^{(\mu)}|(\eta\lambda)^{p+q} = O(1).$$

Therefore, all three SDEs have order 1 weak approximation error.

IV. Capturing escaping behaviour of SGDm near saddle points: momentum SME-2 vs HA-SME

Here we show that when the stepsize is inversely proportional to the Hessian eigenvalues, a situation that can naturally arise locally in practice during deep learning training, momentum SME-2 requires exponentially long time to escape a saddle point, whereas momentum HA-SME escapes in sub-exponential time. It has been observed that for larger momentum SGDm iterates escape quickly a saddle point (Wang et al, Escaping Saddle Points Faster with Stochastic Momentum), so HA-SME accurately captures this rapid escape behavior, aligning closely with that of SGDm, while SME-2 fails to reflect this dynamic.

We follow the proof of Theorem 9 from the HA-SME paper.

Theorem.

Let $0 \le \mu < 1$. Assume that there exists a compact domain $S_0 \subset \mathbb{R}^d$, and let $f : \mathbb{R}^d \to \mathbb{R}$ be a $C^4(S_0)$ function. Suppose that the following conditions hold for all $x \in S_0$:

1. There exists $x_0 \in S_0$ such that $\nabla f(x_0) = 0$, and $\nabla^2 f(x_0)$ has both positive and negative eigenvalues.
2. The Hessian satisfies $\nabla^2 f(x) = \frac{g(x)}{\eta}$, where $g(x)$ is independent of η. Moreover, there exists an absolute constant $c_0 > 0$ such that every negative eigenvalue λ_i of $\nabla^2 f(x_0)$ satisfies $\eta\lambda_i \le -2 - c_0$.
3. The conditions of the well-posedness Theorem in section II.4 hold, so that momentum HA-SME is well-posed.
4. The gradient-noise covariance satisfies $\|\Sigma(x)\| = \Theta(1)$ uniformly on S_0, and $\left|\frac{\partial[\Sigma(x)]_{ij}}{\partial x_k}\right| = O(1)$ uniformly for $1 \le i,j,k \le d$.

Then there exist a compact set $S_1 \subseteq S_0$, with $x_0 \in S_1$, independent of η, and constants $\eta_0, c > 0$ such that, for every $0 < \eta < \eta_0$,

- for momentum HA-SME:

$$\mathbb{E}\big[\tau_{S_1}\big] \le c\eta \log\frac{1}{\eta},$$

- for momentum SME-2:

$$\mathbb{E}[\tau_{S_1}] \geq \exp\left(\frac{c_1}{\eta^2}\right).$$

where

$$\tau_{S_1} := \inf\{t > 0 : X_t \in \partial S_1,\ X_0 = x_0\}.$$

Proof. Set $\widehat{f} := \eta f$. Then $\nabla^2 \widehat{f}(x) = g(x),\ \nabla \widehat{f}(x_0) = 0$. Let $g(x) = U(x)\widehat{\Lambda}(x)U(x)^\top$, where the diagonal entries of $\widehat{\Lambda}(x)$ are denoted by $\widehat{\lambda}_i(x)$. Since $\eta\nabla^2 f(x) = g(x)$, the drift of momentum HA-SME can be written as $b(x) = \frac{1}{\eta}\widehat{b}(x)$, where $\widehat{b}(x)$ is independent of η : $\widehat{b}(x) = U(x)\frac{\log\left(f(\mu I, \widehat{\Lambda}(x))\right)}{\widehat{\Lambda}(x)}U(x)^\top \nabla \widehat{f}(x)$. Similarly, the covariance matrix has the form $\mathcal{D}(x) = \eta\widehat{\mathcal{D}}(x)$, where $\widehat{\mathcal{D}} = U(x)\widehat{S}(x)U(x)^\top$ is independent of η:

$$\widehat{S}_{ij}(x) = [U(x)^\top \Sigma(x) U(x)]_{ij} \frac{\log\Big(f(\mu, \widehat{\lambda}_i(x))f(\mu, \widehat{\lambda}_j(x))\Big)}{f(\mu, \widehat{\lambda}_i(x))f(\mu, \widehat{\lambda}_j(x)) - \dfrac{\mu^2}{f(\mu, \widehat{\lambda}_i(x))f(\mu, \widehat{\lambda}_j(x))} + \mu^2 - 1}.$$

Hence, writing $\widehat{D}(x) := \sqrt{\widehat{\mathcal{D}}(x)}$, the SDE satisfies

$$dX_t = \frac{1}{\eta}\widehat{b}(X_t)\,dt + \sqrt{\eta}\,\widehat{D}(X_t)\,dW_t.$$

We next compute the Jacobian of the rescaled drift at x_0. Since $\nabla \widehat{f}(x_0) = 0$, only differentiation of the gradient factor contributes, and therefore

$$\nabla \widehat{b}(x_0) = U(x_0)\log\big(f(\mu I, \widehat{\Lambda}(x_0))\big)U(x_0)^\top.$$

If $\widehat{\lambda}_i(x_0) < 0$, then since $f(\mu, 0) = 1$ and as shown previously f is strictly decreasing, $f\big(\mu, \widehat{\lambda}_i(x_0)\big) > 1$, and consequently $\log f\big(\mu, \widehat{\lambda}_i(x_0)\big) > 0$. Thus $\nabla \widehat{b}(x_0)$ has positive eigenvalues corresponding to the negative-curvature directions of $\nabla^2 f(x_0)$. In particular, x_0 is an unstable equilibrium of the rescaled drift.

Define the time-rescaled process $Y_t := X_{\eta t}$. Then Y_t satisfies

$$dY_t = \widehat{b}(Y_t)\,dt + \eta\widehat{D}(Y_t)\,d\widetilde{W}_t,$$

where $\widetilde{W}_t$ is a standard Wiener process. The coefficients $\widehat{b}$ and $\widehat{D}$ satisfy the regularity and nondegeneracy conditions from the well-posedness Theorem. Since the Jacobian $\nabla \widehat{b}(x_0)$ has a positive eigenvalue, the exit-time estimate for a small-noise diffusion near an unstable equilibrium implies that there exist a compact set $S_1 \subseteq S_0$, independent of η, and constants $c, \eta_0 > 0$ such that $\mathbb{E}\big[\tau^Y_{S_1}\big] \leq c\log\frac{1}{\eta}$ for every $0 < \eta < \eta_0$, where $\tau^Y_{S_1} := \inf\{t > 0 : Y_t \in \partial S_1,\ Y_0 = x_0\}$. Since $Y_t = X_{\eta t}$, the two hitting times satisfy $\tau_{S_1} = \eta\tau^Y_{S_1}$. Therefore, $\mathbb{E}\big[\tau_{S_1}\big] = \eta\mathbb{E}\big[\tau^Y_{S_1}\big] \leq c\eta\log\frac{1}{\eta}$, which proves the result.

For momentum SME-2, the same rescaling trick and argument from Theorem 9 works, except that we need an additional argument why the eigenvalues of $\nabla^2 \hat{f}(x_0)$ are positive. Indeed, the SDE is:

$$dX_t = -\frac{1}{\eta}\nabla \widetilde{f}_\mu(X_t)\,dt + \sqrt{\frac{\eta\Sigma(X_t)}{1+\mu^2}}\,dW_t, \quad \widetilde{f}_\mu(x) := \frac{1}{1-\mu}\hat{f}(x) + \frac{1+\mu}{4(1-\mu)^3}\left\|\nabla \hat{f}(x)\right\|^2,$$

so we have

$$\nabla^2 \widetilde{f}_\mu(x_0) = \frac{1}{1-\mu}g(x_0) + \frac{1+\mu}{2(1-\mu)^3}g(x_0)^2 = \left(\frac{1}{1-\mu}I + \frac{1+\mu}{2(1-\mu)^3}g(x_0)\right)g(x_0).$$

and if γ_i is an eigenvalue of $g(x_0)$, the corresponding eigenvalue of $\nabla^2 \widetilde{f}_\mu(x_0)$ is

$$\gamma_i\left(\frac{1}{1-\mu} + \frac{1+\mu}{2(1-\mu)^3}\gamma_i\right) > 0$$

since by assumption 2 we have $\gamma_i = \eta\lambda_i(x_0) \leq -2 - c_0 \leq -\frac{2(1-\mu)^2}{1+\mu} - c_0$. □

V. Experiments

1. Error Comparison between the three models

We compare the three models with gradient descent, in both the deterministic and stochastic case, by conducting experiments on a two-hidden-layer neural network with a softplus activation function and cross-entropy loss. The network consists of 10 nodes per hidden layer, and we use the Iris dataset for classification. For SGDm we use Gaussian noise $\Sigma(x) = \sigma^2 I$.

The training starts with the same random initialization for all models, and we compute the objective function value at the first 30 SGDm or 60 GDm iterations and the corresponding continuous timestamp ηk. For GDm we use error $\|X(k\eta) - x_k\|$, while for SGDm we use, similarly to the weak approximation error, $\|\mathbb{E}(f(X(k\eta))) - \mathbb{E}(f(x_k))\|$ (where f is the objective function); the expectations are computed as the average over 100 Monte Carlo runs. In the discretizations of the continuous time approximations of GDm we use Euler stepsize $\frac{\eta}{10}$, while for SGDm we use $\frac{\eta}{100}$.

Part 1. Gradient descent with momentum

Our experiments with values $(\mu, \eta) = (0.3, 0.033)$, $(0.4, 0.022)$ confirm that the momentum PF performs the best, close to but slightly better than the 2nd order flow from Ghosh et al, which has significantly smaller error than the 1st order flow from Kovachki and Stuart. All three models were shown to have $O(\eta)$ error guarantee, but the larger difference in performance of the last two truncated flows is expected because their corresponding piecewise-continuous models have errors $O(\eta), O(\eta^2)$. The slightly better performance of the PF compared to the 2nd order flow agrees with the results from Rosca et al 2023 (figure 7) and is expected because it incorporates the whole underlying series and further truncation (up to order n) of this series in the piecewise-continuous model have error guarantee $O(\eta^{n+1})$, which doesn't make much difference for larger n if the stepsize is small. (Figure 1)

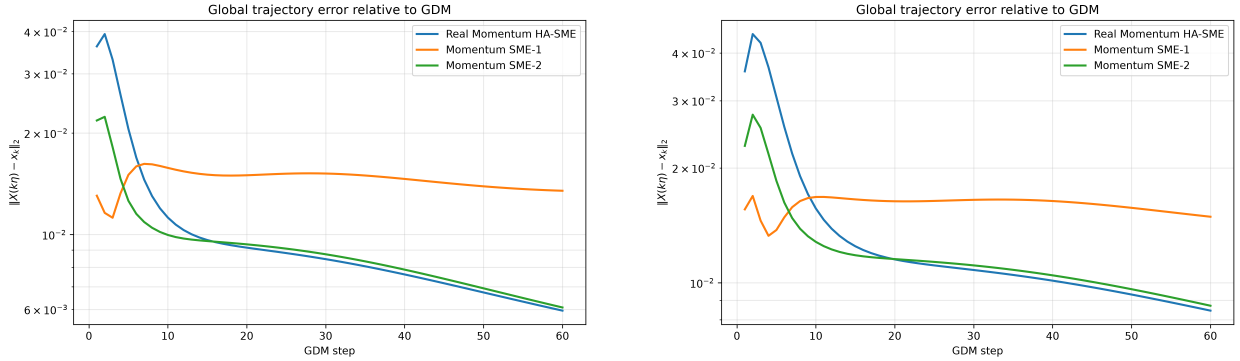


Figure 1: GDm with $(\mu, \eta) = (0.3, 0.033)$ (left) and $(\mu, \eta) = (0.4, 0.022)$ (right)

To further differentiate the models, we conduct another experiment on a neural network with two hidden layers of sizes $10, 1$, sigmoid activation function, $(\mu, \eta) = (0.3, 0.29)$ and Euler stepsize $\frac{\eta}{100}$ (with the other parameters same as before). This confirms more clearly the previous ordering in terms of performance of the three models. (Figure 2)

Part 2. Stochastic Gradient Descent with momentum

For SGDm we get the same ordering of the models for higher number of iterations, but the errors are a bit closer together, which is expected since all models have order 1 weak approximation error. Again, the first order model has the worst performance and the other two are better and very close to each other, which is consistent with the results in Li et al 2024 and the analysis for GDm.

For a bit clearer differentiation between momentum SME-2 and HA-SME, we do another experiment on a neural network with hidden layers of sizes $10, 1$, sigmoid activation function, 40 Monte Carlo runs, $(\mu, \eta, \sigma) = (0.3, 0.29, 0.01)$, 60 iterations, and the other parameters staying the same as before.

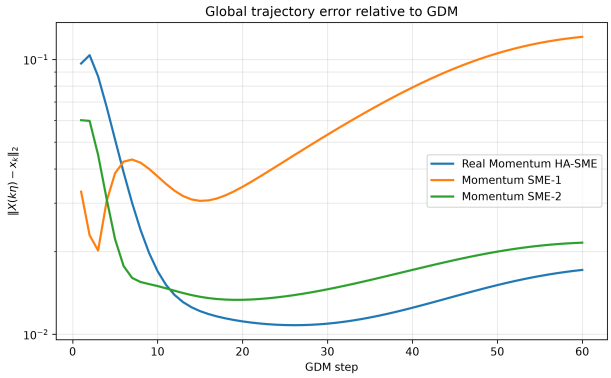


Figure 2: GDm with $(\mu, \eta) = (0.3, 0.29)$

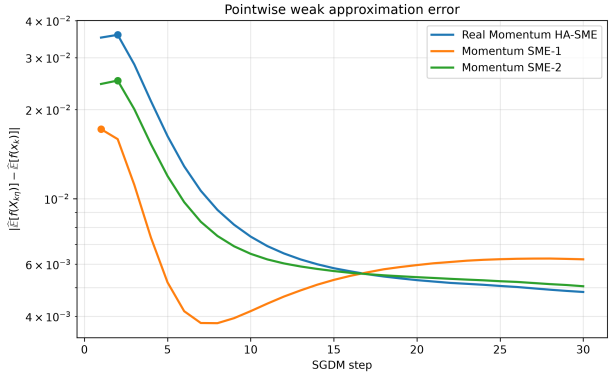


Figure 3: SGDm with $(\mu, \eta, \sigma) = (0.3, 0.033, 0.005)$

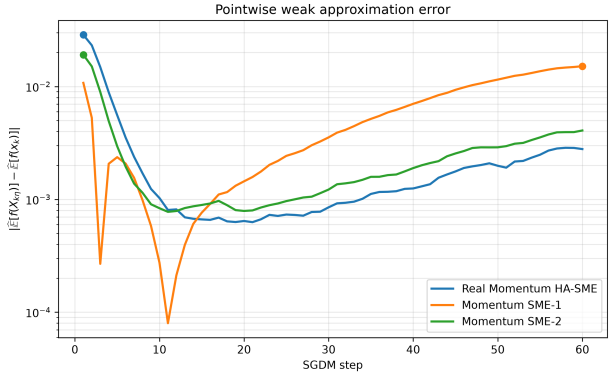


Figure 4: SGDm with $(\mu, \eta, \sigma) = (0.3, 0.29, 0.01)$

2. Stability experiments for GDm

In all experiments below the Euler ratio is 100.

Part 3. Momentum PF in Neural Network Training for larger stepsizes

We compare the discretization errors of the momentum PF and the ODE from Kovachki and Stuart on neural networks for larger stepsizes. We note that the momentum PF is complex for stepsize $\eta > \frac{(1-\sqrt{\mu})^2}{\lambda_0}$ (due to the radii of convergence of the series involved in the momentum PF). As in the stability analysis in I.6, we will consider $\eta > \frac{2(\mu+1)}{\lambda_0}$, where we see PF performs worse than the first order flow (like in Rosca section 4), and $\eta \in (\frac{(1-\sqrt{\mu})^2}{\lambda_0}; \frac{2(\mu+1)}{\lambda_0})$, where we get similar results to the smaller stepsizes (in Part 1). We use a neural network with two hidden layers $10, 1$, tanh activation function, on 60 iterations, $\mu = 0.3$, $\eta = 0.1, 1$. (Figure 5) This confirms Rosca's observation that as the number of iterations increases, the PF accumulates error in the case of $\eta > \frac{2(\mu+1)}{\lambda_0}$.

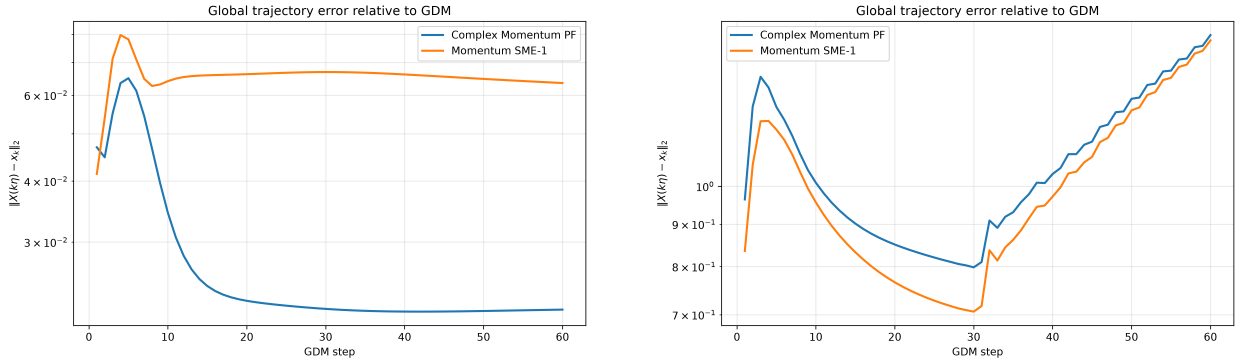


Figure 5: $\eta \in (\frac{(1-\sqrt{\mu})^2}{\lambda_0}; \frac{2(\mu+1)}{\lambda_0})$ (left) and $\eta > \frac{2(\mu+1)}{\lambda_0}$ (right)

Part 4. Continuous-time models of GDm at the edge of stability.

Here we consider the following extension for momentum of the local error used in Rosca's work: instead of $\|x_t - x(\eta; x_{t-1})\|$ and $f(x_t) - f(x(\eta; x_{t-1}))$, we consider $\|x_t + \mu x_{t-2} - x(\eta; x_{t-1}) - \mu x(-\eta; x_{t-1})\|$ and $f(x_t) + \mu f(x_{t-2}) - f(x(\eta; x_{t-1})) - \mu f(x(-\eta; x_{t-1}))$, which correspond exactly to the matching we did in BEA from Section I to derive the PF. Similarly to Rosca section 5, the truncated flows have a larger local error compared to the momentum PF in the edge of stability regime. This means that the momentum PF captures well the gradient descent behavior at the edge of stability, unlike the other two models, which is due to its dependence on the eigendecomposition of the Hessian.

We conducted experiment on a neural network with two hidden layers $10, 10$, activation function tanh, $(\mu, \eta) = (0.2, 0.8)$. The period when λ_0 is around the edge of stability is between steps $10 - 50$ and then clearly the losses of the truncated flows are larger. (Figure 6.)

Part 5. Instabilities in training and stability coefficients.

We observe that the stability coefficients $sc_i = \frac{\log f(\mu, \eta\lambda_i)}{\eta\lambda_i} \nabla f^\top u_i$ derived from the momentum PF (since it's given by $\dot{x} = \sum_{i=0}^{D} sc_i u_i$, similarly to Rosca's proof) are related to the instabilities in training: the increases in loss occur when the corresponding $\mathrm{Re}(sc_0)$ is positive and large. (Figure 7) When $\mathrm{Re}(sc_0) > 0$ GDm resembles the positive gradient flow $\dot{x} = \frac{1}{1-\mu}\nabla f(x)$ rather than the negative one and the positive gradient flow causes instabilities whose strength depends on the stability coefficient sc_0. (Figure 8)

Unlike the edge of stability phase of training, when $\lambda_0 < \frac{2(\mu+1)}{\lambda_0}$ and $\mathrm{Re}(sc_i) \leq 0, \forall i$ and there are no instabilities in the loss and the eigenvalue λ_0 keeps increasing (until it reaches the edge of stability and then we get behavior similar to Figure 7). Moreover, in this early part of training GDm follows closely the negative gradient flow. (Figure 9)

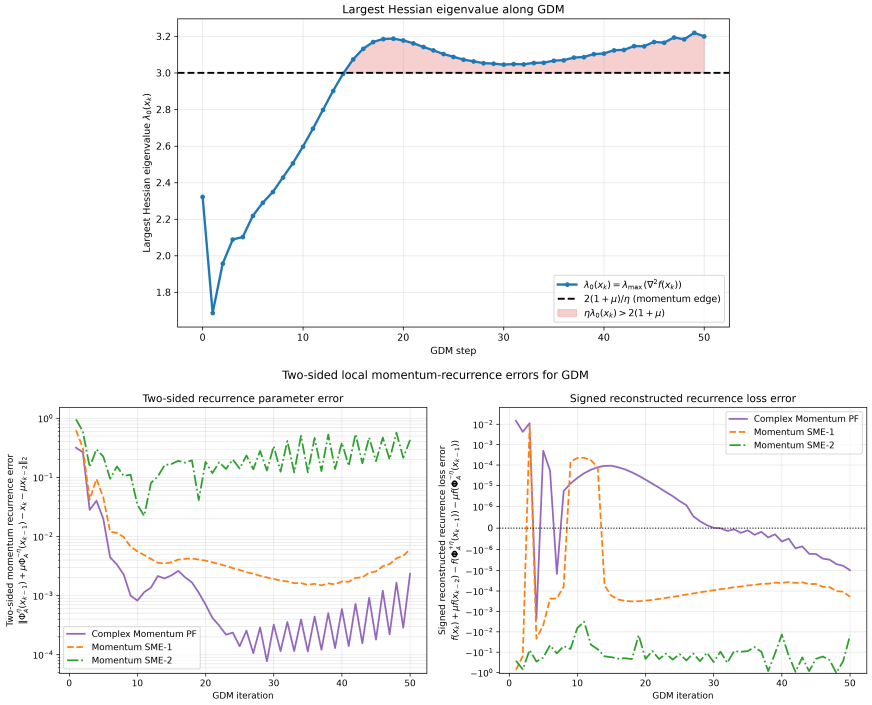


Figure 6: Momentum PF has better local error predictions in the edge of stability regime

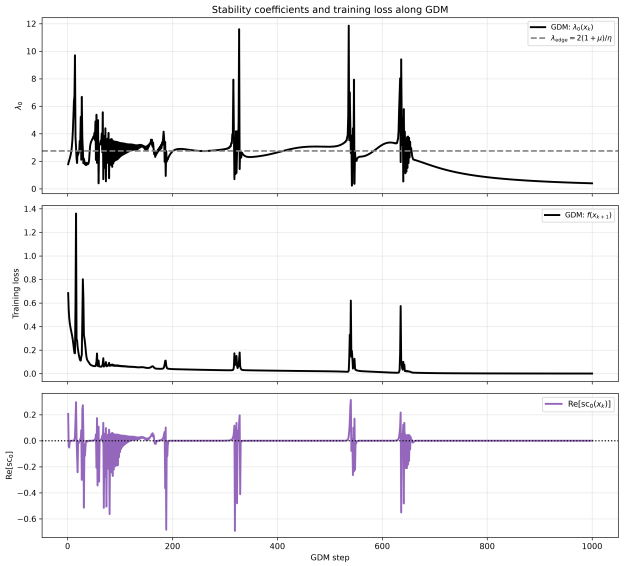


Figure 7: Edge of stability phase of training; NN with 4 hidden layers of size 10 on the Iris dataset, activation tanh, $\mu = 0.1, \eta = 0.8$

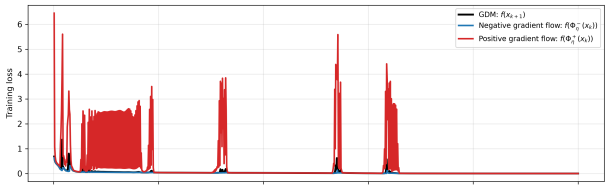


Figure 8: GDm training loss vs. losses of the positive and negative gradient flows for the experiment in Fig 7; GDm resembles the positive gradient flow when $\mathrm{Re}(sc_0) > 0$ is large

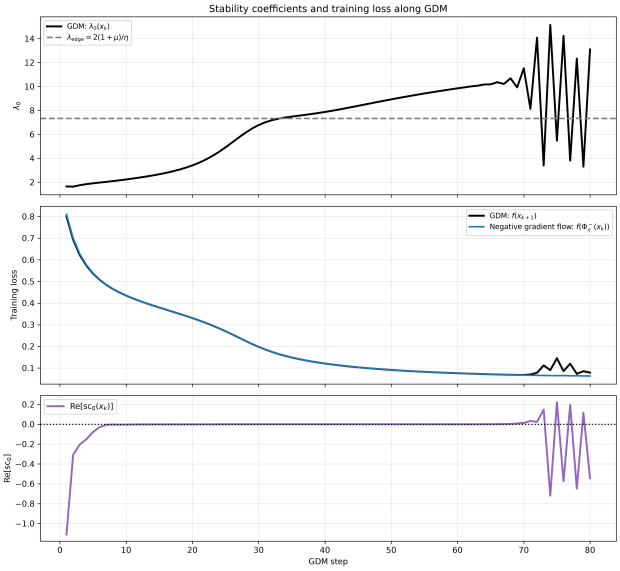


Figure 9: Early training and transition to the edge of stability; NN with 4 hidden layers of size 10 on the Iris dataset, activation tanh, $\mu = 0.1, \eta = 0.3$

Part 6. Stabilizing training by adjusting discretization drift

Like in Rosca's section 6, we note that when $\nabla^2 f \nabla f = 0$, the momentum PF coincides with Kovachki and Stuart's ODE. Indeed, from section I the momentum PF can be written as

$$\dot{x} = -\frac{\nabla f(x)}{1-\mu} + \sum_{n=1}^{\infty} \eta^n C_n (\nabla^2 f)^n \nabla f = -\frac{\nabla f(x)}{1-\mu}.$$

Moreover, this quantity is related to the discretization drift: if we consider the above-mentioned local error as momentum discretization drift (corresponds to the BEA matching and agrees with Rosca's one for $\mu \to 0$), by using Taylor's remainder theorem on $x(t) + \mu x(-t)$(similarly to Theorem 6.1) we have:

$$x_t + \mu x_{t-2} - x(\eta; x_{t-1}) - \mu x(-\eta; x_{t-1}) = -\frac{\eta^2}{2(1-\mu)^2} \left[\nabla^2 f(x(\epsilon))\nabla f(x(\epsilon)) + \mu \nabla^2 f(x(-\epsilon))\nabla f(x(-\epsilon))\right],$$

where $\epsilon \in (0; \eta), \quad x(0) = x_{t-1}$. We can't use this directly because it depends on both $x(\pm\epsilon)$, so we approximate it by another Taylor expansion (and using uniformly bounded gradients):

$$x_t + \mu x_{t-2} - x(\eta; x_{t-1}) - \mu x(-\eta; x_{t-1}) = -\frac{1+\mu}{2(1-\mu)^2}\eta^2 \nabla^2 f(x_{t-1})\nabla f(x_{t-1}) + O(\eta^3).$$

Therefore, we will consider

$$\boxed{\frac{1+\mu}{2(1-\mu)^2}\eta^2 \nabla^2 f(x)\nabla f(x)}$$

as an approximation of discretization drift. Like in Rosca, we define the Drift Adjusted Learning rate so that the signal ($\eta\|g\|$ from the momentum update $x_t + \mu x_{t-2} - (\mu+1)x_{t-1}$) to noise (i.e. the above drift) ratio is 1:

$$\boxed{\eta_{\text{M-DAL}}(x) = \frac{2(1-\mu)^2}{(1+\mu)\|\nabla^2 f(x)\widehat{g}(x)\|}.}$$

Our experiments show that M-DAL significantly improves upon the stability and convergence speed of the training loss compared to using other constant learning rates. Moreover, M-DAL itself is stable and initially decreases in training after which it slowly increases when reaching areas with low drift.

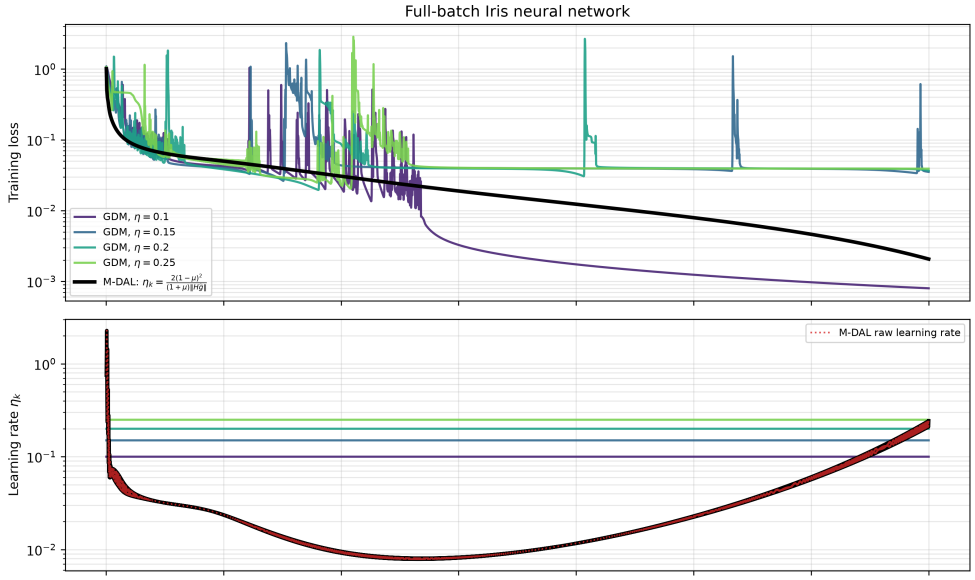


Figure 10: NN with 10 hidden layers of size 10 on the Iris dataset, activation tanh, $\mu = 0.1$